Examples of Confidence Interval - I

In the last module, we showed how the sampling distribution of X bar can be used to compute the probability that X bar will lie within a distance of mu. Now, in Basavaraju's case for instance, since the sample size is 36, which is larger than 30, we can say that regardless of the actual distribution of the satisfaction score in the population. Central Limit Theorem allows us to conclude that the sampling distribution of x bar can be approximated by a normal distribution with a mean corresponding to the population mean mu and a standard error corresponding to sigma x bar given by sigma over root n, where sigma has the historical value of 5 and n is equal to 36. That gives us sigma X bar to be equals to 5 over 6 which is 0. 833. Yes, and because the sampling distribution shows how the values of X bar are distributed around the population mean mu, the sampling distribution of X bar also can give us the probabilities of the difference between X bar and mu.

So, let us take an example from the normal probability template We know that 95 percent of the values of any normal random variable lies within plus or minus 1. 96 standard deviations of the mean. Ashwini and Tejas, do you know how to verify that? I think so, we can look at the standard normal template for instance and enter 0. 95 in the inverse section and we will get a equals to minus 1. 96 and b equals to plus 1. 96. Yes, and if it holds for the standard normal, it holds for any normal distribution.

So now, when the sampling distribution of x bar is normally distributed, 95 percent of the X bar values must lie within plus or minus 1. 96 times the sigma X bar of the population mean mu. In Basavaraju's example, the sampling distribution of X bar is normally distributed with a standard error of 0. 833 as we just saw.

And we have plus or minus 1. 96 sigma X bar to be equal to 1. 96. times 0. 833. And what does that come out to be? That is equal to 1. 63. Yes. So therefore, we can conclude that 95 percent of all x bar values obtained using a sample size of n equal to 36 will lie within plus or minus 1. 63 of the population mean mu.

Now, when we started this discussion, I had said that the general form of an interval estimate of the population mean mu is given by x bar plus or minus some margin of error. So, in Basu

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Raju's case, if we set the margin of error to be equal to 1. 63 and we compute an interval estimate of mu, then we get x bar plus or minus 1. 63. or 57. 31 plus or minus 1. 63. Then we can say that 95 percent of the time the population mean mu will lie within x bar plus or minus 1. 63, which is it lies between 55. 68 and 58. 94. To understand why this is a confidence interval estimate. Let us consider the values of the sample mean X bar that could be obtained if we took three different random samples, each consisting of 36 customers.

Let us say that the first sample mean X bar 1 turns out to have a value as shown in the figure. Then, in this case, we can see that centered at X bar 1, if we form an interval by subtracting 1. 63 to the left. And adding 1. 63 to the right of X bar 1, we will get an interval that includes the population mean mu.

Now consider what happens if we had done a second random sample and obtained a second estimate of the sample mean X bar 2 as shown. Although the sample mean X bar 2 differs from the sample mean X bar 1, we see that the interval with X bar 2 as the center and by subtracting 1. 63 and adding 1. 63 also includes the population mean mu.

However, consider what happens if we had obtained the third sample mean value corresponding to x bar 3. In this case, the interval formed with x bar 3 as a center and by subtracting 1. 63 and adding 1. 63 will not include the population mean mu. This is because x bar 3 falls in the upper tail of the sampling distribution of x bar and it is also further than 1. 63 from mu. Therefore, subtracting or adding 1. 63 to x bar 3 will result in an interval which does not include the population mean. Would see a similar result if the sample mean x bar was instead in the lower tail at a point further than 1. 63 from mu? Yes Tejas, we would if it was a point far away on the lower tail or far away on the upper tail.

Therefore, any sample mean x bar that lies within plus or minus 1. 63 from the population mean mu will create an interval that contains mu. And because 95 percent of all possible sample means must lie within mu plus or minus 1. 63, 95 percent of all such intervals around those x bars obtained by adding or subtracting 1. 63 should contain the population mean mu. And because 95 percent of all intervals constructed using this x bar plus or minus 1. 63 will contain the population mean mu, we say that we are 95 percent confident that the interval from 55. 68

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to 58. 94 includes the population mean mu. We also say that this interval has been established with a 95 percent confidence level.

The value of 0. 95 corresponding to the 95 percent is referred to as the confidence coefficient and often denoted by one minus alpha and the interval 55.68 to 58.94 that we got is called as the 95% confidence interval and the margin of error is given by Z of alpha by two times Sigma X bar or Z of alpha by two times sigma divided by the square root of n.

Definition time. Formally, the confidence interval of population mean mu for the case where the population standard deviation sigma is known is given by x bar plus or minus z alpha by 2 times sigma divided by root n, where 1 minus alpha is the confidence coefficient and z alpha by 2 is the z value corresponding to an area of alpha by 2 in the upper tail of the standard normal probability distribution.

Let us now calculate the 80 percent, the 90 percent and the 99 percent confidence interval for the same population mu for Basavaraju's satisfaction score survey. If you recall what we did for the 95 percent confidence interval, we started with a sample average x bar. which remains at 57. 31 in this case for all confidence intervals of the mean mu.

We have the population standard deviation of 5 and the sample size of n, which also remains the same. So, sigma by root n remains at 5 by 6, which is 0. 833. Then the margin of error given by z alpha by 2 times sigma over root n will require us to only calculate the new values of z alpha by 2 for 1 minus alpha being equal to 80 percent, 90 percent and 99 percent.

Tejus we saw that for the 95 percent confidence interval, 1 minus alpha was 0. 95, the confidence coefficient. What was Z alpha by 2 in that case? It was 1. 96. Okay, then for a confidence coefficient of 0. 99 or 99 percent confidence interval, what would be the Z alpha by 2? In the standard normal template, we enter 0. 99 in the inverse section and we will get A equals to minus 2. 58 and B equals to 2. 58. Therefore, the margin of error would be 2. 58 times sigma divided by root n, which is 2. 58 times 0. 833, which comes out to be equal to 2. 15. This yields a 99 percent confidence interval corresponding to 57. 31 plus or minus 2. 15. That is, we can be confident that 99 percent of the time the population mean will lie between 55. 16 and 59. 46. However, there is a 1 percent chance that it will lie outside this interval. Let us now

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calculate the 90 percent confidence interval. We only need to compute the new value of z alpha by 2 corresponding to the confidence coefficient being 0. 90. Yes, and we get z alpha by 2 to be equal to 1. 64.

Therefore, the margin of error will be 1. 64 times sigma divided by root n, which is 1. 64 times 0. 833, which comes out to be 1. 37. And this yields a 90 percent confidence interval corresponding to 57. 31. plus or minus 1. 37. In other words, we can be confident that 90 percent of the time the population mean will lie between 55. 94 and 58. 68. However, there is a 10 percent chance that it will lie outside this interval.

Okay, let us also calculate the 80 percent confidence interval. We only need to compute the new value of z alpha by 2 corresponding to the confidence coefficient being 0. 8. Yes, and we will get z alpha by 2 to be equal to 1. 28. And therefore, the margin of error will be 1. 28 times sigma by root n, which is 1. 28 times 0. 833, which comes out to be 1.07. This yields an 80 percent confidence interval corresponding to 57. 31 plus or minus 1. 07. That is, we can be confident that 80 percent of the time, the population mean will lie between 56. 24 and 58. 38, but there will be a 20 percent chance that it will lie outside this interval.

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Effect of Alpha and Sample Size on Confidence Interval

So, now that we have computed these various confidence intervals, let us reflect on what we have got. Tejas and Ashwini, what do you notice about the margin of errors as we change the confidence coefficient from 0. 8 to 0. 9 to 0. 95 to 0. 99? We only need to compute the new value of z alpha by 2 corresponding to the confidence coefficient being 0. 90. Yes, and do you find that intuitive? In a sense, yes. If you want to be more confident, the band increases, the interval increases, right? Yes, because we cannot escape the randomness. So, the margin of error corresponding to an 80 percent interval will be smaller than the margin of error corresponding to a 99 percent interval.

So what if we wanted to be a 100 percent sure, a 100 percent confidence interval? Well, let's find out. The confidence coefficient would be 1. Yes, and z alpha by 2 from the normal template would be? Infinity. Yes, and our margin of error Would be infinite. That will give an interval saying mu lies within minus infinity to plus infinity.

Well, even without all the statistical inference, we know that the score has to lie between zero and a hundred. So if you are too greedy, the interval that we get might be pretty less useful. So with confidence intervals. You have to strike a balance when you choose the confidence coefficient. So what is the optimal confidence coefficient? 0. 9 or 0. 95 or 0. 99? Well Tejas, the answer depends on the application. In very critical applications such as FDA approvals for the pharma industry, the FDA might specify that you must use a 99 percent or a 95 percent confidence interval. And in those cases, that is exactly what you do. But in most cases, no one tells you what the confidence coefficient needs to be.

You have to decide that as a decision maker. How do you do that? Well, in a way, you decide that by asking, what is the cost of being wrong? Assuming that mu lies in the interval. This is not a literal cost in most cases. But it's more like an opportunity cost or a risk assessment. So let's say that you are wrong 5 percent of the time in our decision, because we used a 95 percent interval, and the mu could lie outside that interval 5 percent of the time. If the opportunity cost, whatever it is, is pretty high, then you might decide to use a 99 percent confidence interval instead. And by the same logic, you might choose to use a 80 percent confidence interval, as an acceptable level 2 if the cost of being wrong is not very high.

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So what should Basavaraju use? Well, we should ask him, right? Okay. But I think in his case, it is not as strict as a pharma application. So I would think that either a 90 percent or a 95 percent would be okay. And that is the case in most applications. However, it is important not to keep changing this confidence coefficient while comparing different options of decisions.

You just pick a confidence coefficient, whatever is appropriate, and compare decisions using the same confidence coefficients. Don't worry, we will see more of this when we look at some examples.

Okay, we talked about the effect of the confidence coefficient on the interval. Let us now talk about the impact of the sample size on the margin of error. How does the sample size impact the margin of error? Margin of error is z alpha by 2 times sigma divided by root n. Yes, so for a given value of confidence coefficient and alpha by 2, as n increases, the margin of error decreases. Larger the sample size, smaller the margin of error. Yes, and if you have more data, you can be more confident.

Now, when I put it that way, it does seem obvious, doesn't it? But, if you have a small sample, the margin of error might still be acceptable, given the constraints that you had that led to a small sample. Just like we asked what the optimal confidence coefficient is, can we ask what is the optimal sample size? Yes, and the expression for the margin of error is often used to compute a target sample size for a desired margin of error. We will see that a little later.

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Confidence Interval σ (Sigma) known

I want to mention one important point just in case you did not realize about our definition of the confidence interval. If you recall the confidence interval of population mean mu for the case where the population standard deviation sigma is known is given by x bar plus or minus z alpha by 2 times sigma over root n, where 1 minus alpha is the confidence coefficient and z alpha by 2 is the z value corresponding to an area of alpha by 2 in the upper tail of the standard normal probability distribution.

If you are wondering why we use the standard normal z alpha by 2 to compute the margin of error. If you remember, we said that X bar comes from a normal distribution, at least approximately, with mean equal to the population mean mu and standard deviation equal to sigma X bar given by sigma divided by the square root of N.

Therefore, the normal random variable X bar minus its mean mu divided by its standard deviation sigma X bar would be a standard normal. So, if the confidence coefficient 1 minus alpha is 0. 95, then we can say that the standard normal random variable X bar minus mu divided by sigma X bar will take values between minus 1. 96 and plus 1. 96, 95 percent of the time. That is the population mean mu will lie within X bar plus or minus 1. 96 sigma x bar, 95 percent of the time. Yes, thanks for repeating the logic and explaining it. I understand it better now. Can we solve a few examples? Yes, that is exactly what we are going to do next.

In fact, we are going to Kanpur. Amit Mishra works at the famous Thagu Ke Laddu shop in Swaroop Nagar in Kanpur. Thagu ke laddu is a very famous shop, famous for their laddus and badam kulfi. If you ever visit Kanpur, visit one of their outlets, a couple of laddus and a scoop of badam kulfi and your day is made.

Amit has been having a very busy month so far. Business has been unusually high and he is worried that his planning might have to be corrected for the higher level of demand that he is facing. He has collected data and wants to do a quick analysis of the average sales per customer. He has data for 50 randomly selected customers and he finds that the average bill amount is 2200 rupees.

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Historical data says that the standard deviation of the bill is usually 84 rupees. Using this information, Amit would like to compute both the 80 percent and the 95 percent confidence interval for the average bill amount in the population. Can we help, Amit? We have a random sample of 50 customers. So, n is equal to 50. The average bill amount is 2, 200 rupees for the sample. That would be our x bar. Yes, correct. And the population corresponds to all the customers that could visit a shop during a given day. If the sample of 50 is random, and since it is more than 30, we can assume that the distribution of x bar is approximately normal, centered at mu, for which we want the confidence interval, but with a standard error sigma x bar equal to sigma divided by root n.

The standard error then is equal to 84 divided by the square root of 50. Tejas, what does that come out to be? It comes to be 11. 88. Yes, then x bar comes from a normal centered at mu with standard deviation x bar equal to 11. 88. And therefore, we have that x bar minus mu divided by sigma x bar will come from a standard normal.

So now, we compute an 80 percent confidence interval. First, we find the z alpha by 2 corresponding to a confidence coefficient of 0. 8 and that comes out to be 1. 28. Then our margin of error is going to be equal to 1. 28 times sigma x bar, which comes out to be 1. 28 times 11. 88. That comes to be 15. 22. Yes.

So, our 80 percent confidence interval on the population mean mu would be equal to 2200 over X bar plus or minus 15. 22. In other words, mu will lie between 2184. 78 and 2215. 22, 80 percent of the time. Next. To compute the 95 percent confidence interval, we compute the Z alpha by 2 corresponding to the confidence coefficient being equal to 0. 95. The Z alpha by 2 comes out to be 1. 96. Yes, that's right Tejas. So the margin of error would come out to be 1. 96 multiplied by 11. 88, which comes out to be 23. 28. Therefore, 95 percent confidence interval implies that mu will lie within 2200 plus or minus 23. 28. That is between 2176. 72 and 2223. 28, 95 percent of the time.

Let's look at another example. Let's go to Lucknow. Puneet Prasad runs the Swan Spa in Gomti Nagar at Lucknow. Swan Spa is known for its extensive choices of exquisite treatments and massages. It specializes in Ayurvedic, Swedish body massage spa therapy, aromatherapy, body scrubs and foot reflexology massages.

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Puneet takes a sample of ten typical one hour massage therapy sessions and he finds the average to be equal to 3600 rupees. Punit also knows that the population standard deviation is 500 rupees. What assumptions regarding the population should we make in order to compute the confidence intervals? And he wants to know at 95 percent confidence what is the margin of error and at 99 percent confidence what will be the margin of error.

Tejas, let us get started. What assumptions do we need to make to compute these confidence intervals? The sample size is small, in this case, less than 30. So, can we approximate the distribution of X bar to be normal? No, we cannot. But if we assume the population to be normal, X bar will have a normal distribution.

Yes, Ashwini, you are right. If we assume the population to be normal, then X bar will already have a normal distribution. So, we need to assume that the amount customers spend on massages at Swan Spa has a normal distribution centered at mu with a standard deviation of 500 in the population. Then X bar, the average of 10 random samples from this population will also be a normal random variable centered at the mu, but with a standard deviation given by sigma x bar, which is equal to given by sigma divided by root n, which is 500 divided by root 10, which is 158. 11 rupees.

Yes, and therefore, we will have that x bar minus mu divided by sigma x bar will come from a standard normal distribution. So, now we compute the 95 percent confidence interval. First, we find the z alpha by 2 corresponding to a confidence coefficient of 0. 95 and that comes out to be equal to 1.96. So, then our margin of error is going to be equal to 1. 96 times sigma x bar which comes out to be 1. 96 times 158. 11. Then our margin of error is 309. 90. Yes, so our 95 percent confidence interval on mu would be equal to 3600 plus or minus 309. 90. That is mu would lie between 3290. 1 and 3909, 90. 95 percent of the time. Next, to compute the 99 percent confidence interval, we compute Z alpha by 2 corresponding to the confidence coefficient being equal to 0. 99.

The Z alpha by 2 comes out to be 2. 58. Yes, that's right. So the margin of error comes out to be 2. 58 multiplied by 158. 11, which comes out to be 407. 27. Therefore, the 99% confidence interval implies that new will lie within 3,600 plus or minus 407.27. That is between 3 1 9, 2 0.73 and 4 0 0 7 0.27 99% of the time.

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We see how the margin of error increases from 309 to four seven as we go from 95% to 99%. Yes, and do you think our normal assumption is reasonable here? How would we verify? Like Dr. Sajdev, we could plot the sample data and see. Why cannot he just use a larger sample? There will not be much cost or time involved.

Yes, that's a good point, Ashwini. And in fact, we should just tell him that.

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Confidence Interval - σ (Sigma) Unknown

We look at developing a confidence interval for the population mean when sigma or the standard deviation of the population is unknown. In fact, this is a more common case you would agree because when you are developing a confidence interval estimate for a population mean we usually do not have a good estimate for the population standard deviation either.

In such a case, we must use the same sample to estimate both the mean mu and the standard deviation sigma. This situation is commonly referred to in textbook as the sigma unknown case. So when the sigma of the population is unknown, mu of the population is unknown and we want a confidence interval on this mu and we have only one sample. The natural alternative is to use the sample standard deviation, S, instead of the population standard deviation, sigma. Does this just involve replacing sigma in the margin of error equation with S and using the normal template as we did? No, Tejas, no. Unfortunately, we need to do a little bit more. When we use S, the estimate of the standard deviation from the sample, instead of the population sigma, then the confidence interval and the margin of error for the population mean mu, requires the use of the student t distribution instead of the standard normal distribution.

Tejas and Ashwini, do you remember the student t distribution? Yes, I do remember, but why should we use the t distribution, why can't we just use z? Yes, let us see why. Recall that when sigma was known, we had argued that X bar comes from a normal distribution and therefore, X bar minus mu divided by sigma over root n was a standard normal random variable.

Now, in this case, we do not know sigma and we decide to use the sample standard deviation instead of sigma in the denominator. Now, even though we do not know sigma, X bar is still a normal random variable and it is still centered at mu. And the standard error is given by sigma divided by root n. So, x bar minus mu will be a normal shifted by mu. So, it is normal still. But when we divide by s over root n instead, we are no longer dividing by standard deviation anymore. Oh, so it is not a standard normal any longer, but a different normal. No, x bar minus mu divided by s over root n. It is a T random variable with n minus 1 degrees of freedom.

Realize that the S in the denominator is also a random variable, since it comes from the sample. Oh, that's right. So now we are dividing one random variable X bar minus mu by another random variable S over root n. Yes, exactly. And if

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you recall, this was the definition of the student t distribution with n minus 1 degrees of freedom, where n is the sample size.

Now, technically, this is true only when our sample is drawn from a normal population. Now, Tejas and Ashwini, what do you remember about the student t distribution? It was founded by William Gosset during his internship at Guinness. That is the first thing I remember. Yes. And any mathematical properties?

It is bell shaped, centered at zero, like Z, and symmetric. Yes. Excellent. What else? It has degrees of freedom of n, and it is flatter than Z. Yes. Very good. Anything else? As the degrees of freedom increases, it starts to converge to the standard normal. Very good. So, therefore, for the sigma unknown case, we can compute the critical values for the t distribution just like we computed the critical values for the standard normal distribution.

So, although you were disappointed that when we use s, suddenly new complications arose and we could no longer use the standard normal. The alternative of using the student t distribution is definitely doable and not very different from using the Z as the t distribution looks a lot like the Z. Although the mathematical development of the t distribution is based on the assumption of a normal distribution for the population, in many cases the t distribution can be used even if the population distribution deviates slightly from the normal.

Later on, I will talk about some of the guidelines for using the t distribution in more detail. But now that we have realized that we need to use the t distribution, let us first see how to use the t distribution to compute the margin of error and the confidence interval for the case where sigma is unknown.

For the sigma known case, we saw that the interval estimate of a population mean is is given by x bar plus or minus z alpha by 2 times sigma divided by root n. To compute the interval estimate for mu for the sigma unknown case, the sample standard deviation s is used to estimate sigma and z alpha by 2 is replaced by the t distribution value t alpha by 2 corresponding to a t distribution with n minus 1 degrees of freedom where n is the sample size.

The margin of error is then given by T alpha by 2 n minus 1 times s divided by root n. With this margin of error, the general expression for an interval estimate can be written. So let us now see the formal definition. Definition term. The interval estimate of a population mean for the sigma unknown case is given by x

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bar plus or minus t alpha by 2 n minus 1 times s over root n, where s is a sample standard deviation and 1 minus alpha is the confidence coefficient. And t alpha by 2 n minus 1 is the t value corresponding to an area of alpha by 2 in the upper tail of a t distribution with n minus 1 degrees of freedom.

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Examples of Confidence Interval - σ (Sigma) Unknown

Let us now apply this learning to a few examples. Let us revisit an example that we saw earlier. If you recall, Joti Hegde was the head of Placement and alumni relations at IIT Dharwad. She had a survey of 103 respondents, and in her survey she had asked them several questions, including salary information, satisfaction with the program and performance on various test scores.

When we discussed descriptive statistics in module 1, we had looked at the salary information and computed various descriptive statistics for the salary of the respondents. Here is the data that we had collected. In this case, we do not know the mean or the standard deviation of the population. However, using this sample data and its summary statistics and what we have learnt recently, we can establish confidence intervals for the salary.

Let us establish confidence intervals for the confidence coefficient of 0. 9 and 0. 95. First, we note that the sample size n is equal to 103 in this case. And we have the average salary from the sample X bar to be equal to 25. 92 lakhs. The sample standard deviation S is equal to 3. 66 lakhs. Then the 90 percent confidence interval of the unknown population mean mu can be computed as X bar plus or minus T alpha by 2 N minus 1 times S over root N. Now, S is the sample standard deviation, which is 3. 66 in this case, and n is 103. Therefore, S over root n will be given by, it comes to 0. 361. Yes, exactly. So, now, we need to compute the T alpha by 2 n minus 1. from a t distribution with 103 minus 1 or 102 degrees of freedom for an alpha by 2 corresponding to 0. 05.

Ashwini, can you check our t template and find out what that comes out to be? In the t template, first we update the degrees of freedom and we go to the inverse section and find the value corresponding to a tail probability of 0. 05 as 1. 66. Yes, nice work. So now that we have what we need to compute the margin of error, let's compute it.

The margin of error is T alpha by 2 n minus 1 times s over root n. And what does that come out to be? It is 1. 66 times 0. 361, which is 0. 60. Yes, and so therefore, the 90 percent confidence interval on the average population salary would be x bar plus or minus the margin

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of error, which is 25. 92 plus or minus 0. 6 lakhs. That is, it lies in the interval 25. 32 to 26. 52 lakhs.

Next, let us compute a 95 percent confidence interval. What changes? Only margin of error, as we have a new value of alpha by 2. Yes, that is right. We will need a new value of T alpha by 2 n minus 1. Now, alpha by 2 is 0. 025 and the degree of freedom is still equal to 102, which is 103 minus 1.

So, the value of T alpha by 2 n minus 1 is going to be equal to 1. 98 and the margin of error would then come out to be. 1. 98 times 0. 361, which is equal to 0.72 therefore, the 95 percent confidence interval on the average population salary is x bar plus or minus the margin of error, which would come out to be 25. 92 plus or minus 0. 72 lakhs. That is, it lies in the interval from 25. 2 to 26. 64 lakhs. This is pretty much like what we did for the sigma known case, except with the t distribution. Yes, it is and we need to remember that we need a t n minus 1 when the sample size is n for the confidence intervals. Now, is that always the case? Yes, always.

Now, let us do another example. We go back to Hanumantha Pai with Canara Bank. Do you remember Hanumantha Pai of Canara Bank? Vaguely. I need to check the notes. Let me do that. Yes, go ahead. Look at your notes. Yes, found it. He had 55 data points in his survey and was looking at amount charged to credit cards in one month. Yes, and the average amount spent was from the sample equal to x bar of 3852. 11. And the sample standard deviation S was equal to 1454. 23 and the sample size that he had was 55.

Let us now compute the 80 percent confidence interval and the 95 percent confidence interval for the average spend in the population for Hanumantha. Then, the 80 percent confidence interval on the unknown population mean mu can be computed as x bar plus or minus t alpha by 2 n minus 1 times s over root n. Now, s is the sample standard deviation which is 1454. 23 in this case and n is 55. Therefore, the s over root n would be given by, it comes to 196. 09.

Next, we need to compute the T alpha by 2 n minus 1 from a T distribution with 55 minus 1 or 54 degrees of freedom for an alpha by 2 corresponding to 0. 1. Ashwini, can you check from our T template what that comes out to be? In the t template, first we update the degrees of freedom to 54 and we go to the inverse section and find the value corresponding to a tail

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probability of 0. 1 as 1. 3. Then the margin of error is given by t alpha by 2 n minus 1 times s over root n. And what does that come out to be, Tejas? It is 1. 3 times 196. 09, which is 254.41. So therefore the 80% confidence interval on the average monthly spend in the population would be X bar, plus or minus the margin of error, which would be 3852.11, plus or minus 254.41. That is, it would lie in the interval, 3597.7 to 4106.52.

Next, let us compute a 95 percent confidence interval. What changes? Only margin of error as we have a new value of alpha by 2. Yes, that is right. We will need a new value of T alpha by 2 n minus 1. Now alpha by 2 is equal to 0. 025 and the degree of freedom is still equal to 54 which is 55 minus 1.

So the value of T alpha by 2 n minus 1 is equal to 2. And the margin of error is 2 times 196. 09, which is equal to 393. 13. Therefore, the 95 percent confidence interval on the average population salary will be X bar plus or minus the margin of error, which would be 3852. 11 plus or minus 393. 13. That is, it would lie in the interval 3458.98 to 4245. 24. Quite straightforward. I like it. Yes, it is. Notice how the margin of error is bigger when you go from an 80 percent to a 95 percent confidence interval. Yes, it does get wider. Also, when I first saw the t distribution, I thought it will be painful when we use it. Now it is just as easy as the standard normal.

Yes, that is the case and the t distribution does have good property. We have seen how to compute the confidence intervals for both the sigma known and the sigma unknown case. Let us talk a little bit about how they are used in practice and their reliability. If the population follows the normal distribution, then the confidence interval and the margin of error provided by the expression that is based on the t distribution is exact and can be used for any sample size from the population. If the population does not follow the normal distribution, then the confidence interval and the margin of error provided by the expression that is based on the t distribution will be an approximation. In this case, the quality of the approximation will depend on both the distribution in the population and how large of a sample we have obtained.

In most cases, if you have a sample of 30 or more, then, it is more than adequate to use the equation that involves the t distribution to develop the interval estimate for the population mean. However, if the population distribution is highly skewed or contains outliers, for

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example, as in the case of Dr. Sazdev that we saw earlier, then it would be better to increase the sample size to 50 or more. And if you recall, we had suggested this at that time. But if the population is not normally distributed but is roughly symmetric or bell shaped, then even a small sample size as small as 15 can give us good approximations for the confidence intervals.

For small sample sizes, the expression for margin of error computation based on the t distribution should only be used if we are okay to assume that the population distribution is approximately normal, even if it is known to be not exactly normal. We have seen two approaches for developing the confidence interval estimate for the population mean mu.

For the sigma known case, the sigma and the standard normal distribution are used to compute the margin of error and to develop the confidence interval estimate. For the sigma unknown case the sample standard deviation and the t distribution are used to compute the margin of error and develop the confidence interval estimate.

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Determining the Sample Size

Let us now look at how to compute the sample size using what we have learnt about confidence intervals and margin of errors. As we have been discussing confidence intervals, we have talked about the role of the sample size n in providing good and approximate confidence intervals when the population is not normally distributed.

Next, we will focus on another aspect related to the sample size. We will talk about how to choose a sample size that is large enough to provide a desired margin of error. Realize that the sample size decision is a decision that is taken prior to collecting the sample data rather than after the data collection.

So how does this work? Let us look at the case where sigma is known. If you remember, we had said that the confidence interval for the mean mu is written as x bar plus or minus z alpha by 2 times sigma divided by root n, where the quantity of z alpha by 2 times sigma divided by root n was the margin of error.

We see that the term z alpha by 2, the population standard deviation and the sample size combined together determine this margin of error. Now, let's say that we have selected a confidence coefficient or one minus alpha, then Z alpha by two can be determined using this confidence coefficient. If you have an estimate of the value of sigma, also, then we can determine the sample size and that will provide us a desired margin of error.

We compute this as follows, let E denote the desired margin of error. Then, this E is equal to Z alpha by 2 times sigma divided by the root of N. So, the value of the sample size N that would yield the desired margin of error would be given by the square of Z alpha by 2 times sigma divided by E. But to compute the sample size, we would need the population sigma and we may not even know a sample estimate of s before we do a survey. How does this work then? That's a good question Tejas. So, we can do a few things. First, we can use an estimate of the population standard deviation computed from previous studies as a planning value for sigma. If you recall, we had Basavaraju use a value of sigma of 5 from the last year. Second, you could conduct a pilot study to get a preliminary sample and we can use the sample standard deviation from this preliminary sample as a planning value of sigma.

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Third, we could use our judgment or a best guess for the value of sigma. For example, we could find the largest and the smallest value in the population and the difference between the largest and the smallest value that is the range. And then divide that range by 4 and use that as a rough or crude approximation for the standard deviation.

Let us now look at an example that demonstrates how this can be used to compute the sample size. Let's go back to the case of Basavaraju 's customer satisfaction survey. If you recall, he had found a sample average of 57. 31 using a sample size of n equal to 36. He had a historical value of sigma for the population that was equal to 5.

We had used this value to compute the confidence intervals. In fact, we had computed a 99 percent confidence interval for his population mean. Tejas, can you pull out the margin of error that we got then? Yes. In my notes, I have that margin of error as plus or minus 2. 15. giving us a confidence interval of 57. 31 plus or minus 2. 15. Yes, so we get the interval to be in the range of 55. 16 to 59. 46. Now, let us say that Baswaraju wants to establish a 99 percent confidence interval again, but now with a margin of error of plus or minus 1. In this case E, the desired margin of error would be 1. What should have been the sample size that Basavaraju should have collected?

So that on subsequent analysis, the margin of error for 99 percent confidence interval would have been plus or minus 1. We saw that the confidence coefficient of 0. 99 gives us a z alpha by 2 of 2. 58. We also know that the sigma of the population is 5 and e, the desired margin of error is equal to one. So the required sample size that Basavaraju should have used is given by the square of Z alpha by two times and sigma divided by E.

And what does that come out to be Ashwini? In this case, it would be the square of 2.58, multiply by 5 divided by one, which is 165.8. Yes. So if Basavaraju had used a sample of 166, then we can say that on analysis of such a sample, he would have been able to establish a 99 percent confidence interval with a margin of error of plus or minus 1.

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Confidence Interval for Proportion

Next, we are going to discuss how to construct confidence intervals for the population proportion p. When we started this module, we had said that the general form of the interval estimate for population proportion p would be to write it as a sample estimate p bar plus or minus a margin of error. Of course, The sampling distribution of P bar will be important in determining this margin of error.

What do we know about the sampling distribution of P bar? P bar was defined as x over n, where x is the proportion of the sample that satisfied our success criteria. Yes, and so P bar was defined as a binomial random variable. Then, we said that if our sample size n was large enough, This binomial could be approximated by a normal.

What criteria did we want to be satisfied? Was it n greater than 30? Not exactly, something more specific. We wanted so that n times p is greater or equal to 5 and n times q is also greater or equal to 5. Yes, that is what we wanted, where n is the sample size and P and Q are the probability of success and failure.

And what did we know about this normal distribution that approximate the distribution of P bar? We know that it is centered at the unknown population parameter P and has a standard error sigma P bar defined as the square root of P times 1 minus P divided by n. Now, because the sampling distribution of p bar is normally distributed, like the distribution of x bar, we can use a similar approach to write the margin of error.

We can say that the margin of error required for the confidence interval of proportion is given by z alpha by 2 times sigma p bar. Setting the margin of error in this manner would allow us to construct a confidence interval with 1 minus alpha as the confidence coefficient. We can say that 1 minus alpha fraction of the intervals generated using this value of the margin of error will contain the true population proportion.

This is similar to what we did for sample mean mu using x bar. Yes, the same approach carries forward except there we had a case where sigma was known and sigma was not known. Oh, well we have a similar problem here, don't we? Yes, and what is the problem? The sigma p bar,

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which is given by square root of p times 1 minus p divided by n has p in it, and we don't know p.

Yes, in fact we are trying to construct an interval for p. So, what do we do? You state a solution. No, actually it is simpler in this case. Since sigma p bar cannot be directly used in the computation of the margin of error as we do not know p, we use an approximation for sigma p bar, where instead of p, we use p bar in the equation for standard error.

So, the margin of error for the population proportion is computed as Z alpha by 2 multiplied by the square root of p bar times 1 minus p bar divided by n. Now, with this margin of error, we can write the formal definition and expression for the confidence interval of a population proportion. Definition time, the confidence interval estimate of a population proportion p is given by the p bar plus or minus z alpha by 2 times the square root of p bar times 1 minus p bar divided by n, where 1 minus alpha is the confidence coefficient and z alpha by 2 is the z value corresponding to an area of alpha by 2 in the upper tail of a standard normal distribution.

Let us see how this works using Basavaraju's Sample Survey of Customer Satisfaction. If you recall, we had estimated a sample proportion P bar. Do you remember what that was? Yes, that was the proportion of customers that had given a satisfaction score of 60 or above. Yes, Basavaraju had defined success to be the responses that gave a score of 60 or more.

And out of the 36 responses, there were 17 responses that had given a satisfaction score of 60 or more. And what p bar did it give us? That gave us the sampling proportion p bar equals to 17 by 36 or 0. 47. True. So the question is, how can we establish a confidence interval on the unknown population parameter p using the sample estimate p bar? Let's say we want a confidence coefficient of 1 minus alpha to be equal to 0. 95, so that we can compute a 95 percent confidence interval. Then we know that the margin of error is given by z alpha by 2 times the square root of p bar times 1 minus p bar divided by n. Let us first compute z alpha by 2.

Since the confidence coefficient is 0. 95, we need to find the value of z corresponding to the tail probability of 0. 025. That comes out to be equal to 1. 96. Next, we compute the square

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root of p bar times 1 minus p bar divided by n. With p bar equal to 0. 47, we need the square root of 0. 47 times 0. 53 divided by 36.

And what does that come out to be? It ages. That comes out to be equal to 0. 08. Yes. And so therefore, the margin of error is computed as 1. 96 times 0. 08, which comes out to be equal to 0. 16. Then the confidence interval can be written as p bar plus or minus this margin of error, that is 0. 47 plus or minus 0. 16 which would imply that the population parameter P will lie in the interval of 0. 31 to 0.64, 95 percent of the time.

Let us look at one more example on establishing the confidence interval for population proportion. Let us look at Hanumantha's data set corresponding to the monthly credit card spend of 55 randomly chosen customers. We saw earlier that the sample average was 3582 rupees and the sample standard deviation was 1454 rupees.

Using this information, we had established the confidence interval for the mean monthly spend in the population using the t distribution. Now Hanumantha is interested in the proportion of customers that spend less than 3000 per month. In the given sample of 55 customers in total, there are 16 such respondents. So, what is the P bar? Here, X, the number of customers spending less than 3000 is 16. And so P bar is 16 divided by 55 which is 0. 29. Correct. And so, the confidence intervals will tell us what we can infer from this sample statistics of 0. 29 about what is the proportion that is likely to be there in the population.

Hanumantha wants to establish an 80 percent confidence interval and a 99 percent confidence interval. So, Tejas let us first compute the 80 percent confidence interval that is with a confidence coefficient of 0. 8. We know that the margin of error is given by z alpha by 2 times the square root of p bar times 1 minus p bar divided by n. We should first compute z alpha by 2. Yes, and since the confidence coefficient is 0. 8, we need to find that value of z corresponding to a tail probability of 0. 1 and that comes out to be equal to 1. 28. Next, we divide by the square root of p bar times 1 minus p bar divided by n. With p bar equal to 0. 29, we need the square root of 0. 29 times 0. 71 divided by 55. And what do we get, Ashwini? We get 0. 06. Correct. And so therefore, the margin of error can be computed as 1. 28 times 0. 06, which comes out to be equal to 0. 08. Then. So, the confidence interval can be written as p bar plus

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or minus this margin of error that is 0. 29 plus or minus 0. 08, which would imply that the population parameter p will lie in the interval 0. 21 to 0. 37, 80 percent of the time, good.

Now, what if we wanted a 99 percent confidence interval? We compute a new value of z alpha by 2 corresponding to the confidence coefficient of 0. 99. Yes, so corresponding to a tail probability of 0. 005, the z value comes out to be equal to 2. 58. What happens to the margin of error then? The margin of error is 0. 06 times 2. 58, which is 0.16. Yes, and so there is a 99 percent chance that the population proportionally will lie in the interval 0. 29 plus or minus 0. 16, that is between 0. 13 and 0. 45 are we doing this correctly? Yes, but why do you ask Tejas? These intervals were fairly wide. 99 percent of the time it lies between 0. 13 and 0. 45. How is Hanumantha going to use such a wide interval to make decisions? Yes, and that is a valid point. Ashwini, any idea what is happening? What kind of range would you want for these intervals?

I had not realized this, but now that Tejas is pointing it out, these intervals seem too wide to be useful. I would want intervals that are plus or minus 0. 05. Yes, Tejas and Ashwini, let us build on your questions here and discuss about sample sizes that would lead to a desired margin of error. To answer our questions, let us consider how large should the sample size be so that Hanumantha can obtain an estimate of the population proportion that is within a desired level of precision. For example, the 0. 05 that Ashwini just mentioned. And we can follow an approach for sample size determination that is similar to the approach that we used for the sample size determination for the mean. Earlier, we had said that the margin of error associated with an interval estimate of the population proportion is given by z alpha by 2 times the square root of p bar times 1 minus p bar divided by n. As we saw with the case of the population mean, the margin of error for population proportion is also based on three components. The Z alpha by 2, the sample proportion p bar and the sample size n. Like in the case of sample mean, a large sample size will provide a small margin of error.

Let E denote the desired margin of error related to the population proportion, like the 0. 05 that Ashwini suggested. Then we know that this margin of error E is equal to z alpha by 2 times the square root of p bar times 1 minus p bar divided by n. And like we did for the population mean. We solve for n from this expression and we get the sample size of n that would yield this desired

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margin of error is given by z alpha by 2 square times p bar times 1 minus p bar divided by e square.

Here again we have a problem like we did with the sample size estimation for the mean. Yes Tejas, you are right. We cannot use this formula to compute a sample size that will provide the desired margin of error directly because this expression requires p bar and p bar will not be known until and after we select a sample.

So therefore, what we need then is a planning value of p bar that can be used to compute the desired sample size. How do we get this planning values of p bar? Well, let's say we call this planning value of p bar as some p star. Do we use some approximations like we did in the case of the mean? Yes, Ashwini, that is exactly what we do.

In fact, the planning value of p star can be chosen using a few methods. First, we can use the sample proportion from a previous sample study that Hanumantha might have done in the past. Second, we could do a pilot study, select a preliminary sample and then use the same sample proportion from this pilot study as the P star.

Third, we could use some judgment or best guess for the value of P star. If none of these alternatives work, we could use a planning value of p star that is simply equal to 0. 5. So the approach is pretty much similar to what we did for the mean. Yes, that's right. And so the required sample size that yields a desired margin of error e can be estimated using p star determined by any of these methods instead of the p bar in the expression that we just discussed for e.

Now let's apply this to answer your question about Hanumantha Survey. Ashwini, you had said that you would have preferred that the margin of 0.05. Let us say that Hanumantha agrees with you and wants a 99 percent confidence interval estimate for the population proportion with a desired margin of error of 0. 05. Currently, for the 99 percent confidence interval, the margin of error is 0. 16. And we want to find the sample size that will reduce this margin of error from 0. 16 to 0. 05. Now, since we are still focused on a confidence coefficient of 0. 99, the value of Z alpha by 2 remains at 2. 58. Now, we need a planning value of P star to answer the sample size question. Let us use a planning value of P star equals to 0. 5. Okay, let's go with that. Then

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the desired sample size n would be given by z alpha by 2 square times P star times 1 minus P star divided by e square, which would come out to be 2. 58 square times 0. 5 square divided by 0. 05 and that comes out to be equal to 665. 64 or 666 roughly. Oh, that is a lot higher than what Hanumantha collected from 55 to 666.

Yes, so instead of relying on just about 50 samples or so that Hanumantha had pulled out, If he had pulled out a sample of 666 or more at random, then he would have been able to get a much better margin of error for his 99 percent confidence interval.

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Confidence Interval for Variance

We have talked about confidence intervals for population mean mu and population proportion p. Next, we talk about the confidence interval for the population variance sigma square. In determining the confidence interval for mean, the sampling distribution of X bar was vital. Similarly, for the confidence interval of proportion The sampling distribution of p bar was essential.

For the confidence intervals for the population variance sigma square, the sampling distribution of the quantity n minus 1 times s square divided by sigma square is going to be important. We saw this quantity when we talked about the sampling distributions in general in the last module. What do you remember about the distribution of this quantity?

It is a random variable, since it involves s square from the sample in it. Yes, of course. And if you remember, we made some assumptions about the normal population for variance in the last module. Can you check your notes? Yes, if we are sampling from a normal population, then it is a chi square random variable.

Yes, if you are sampling from a normal distribution, then the random variable n minus 1 times s square divided by sigma square has a chi square distribution with n minus 1 degrees of freedom. And we will use this information to develop the confidence interval estimates for the population variance sigma square.

Let's see how this is done using an example. Let's look at Basavaraju's customer satisfaction survey again. The sample size was 36, that is n is equal to 36, x bar was 57. 31, and the sample standard deviation was 6. 38, and the sample variance was equal to 40. 7. Suppose we want to use this information to develop a 95 percent confidence interval estimate for the population variance sigma square.

Therefore, we have the confidence coefficient to be equal to 0. 95. Now, unlike the case for population mean and population proportion, where we wrote the confidence interval as a sample estimate plus or minus a margin of error for population variance, we will write it directly stating that the population variance lies between a lower value and an upper value 95

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percent of the time.How do we estimate this upper and lower values? To find these upper and lower values, we need to see some details of a chi square random variable. Let chi square alpha denote the value of a chi square random variable such that the area to the right under the PDF, that is, the tail probability is equal to alpha.

Now, Tejas and Ashwini, do you remember what we call such values? We called them the critical values. We calculated them for z and t distribution. Yes, exactly. And so, now that the confidence coefficient for Basavaraj's example is 0. 95, let us look at a chi square with 36 minus 1, that is 35 degrees of freedom.

And let's find the values of chi square alpha corresponding to alpha to be equal to 0.025 and alpha equal to 0. 975. Can you both find these values? Pull out your chi square template. Yes, I have the template open. Put the degrees of freedom to be n minus 1 equals to 35. Yes, and then if you use the inverse section to find the corresponding values for 0. 975 and 0. 025. For tail probability of 0. 025, I get 53. 20. For tail probability of 0.975, I get 20.57. Good. So, 2. 5 percent of the time, the chi square of 35 will take values above 53. 20 and 97. 5 percent of the time, it will take values more than 20. 57. This means that 95 percent of the time a chi square 35 will take values between 20. 57 and 53. 2.

Now, since n minus 1 times s square divided by sigma square is a chi square 35, we have the following inequality that 20. 57 is less than or equal to n minus 1 times s square divided by sigma square is less than or equal to 53. 20. Now, we do some algebra to get our confidence interval. First, we take reciprocals throughout. So, we must switch the direction of the inequalities. Then, we multiply throughout by n minus 1 times s square. And if we do that, we get the following. n minus 1 times s square divided by 53. 20 is less than or equal to sigma square is less than or equal to n minus 1 times s square divided by 20. 57.

Substituting n is equal to 36 and s square is equal to 40. 7, we get that the 95 percent of the time 26. 79 is less than or equal to sigma square is less than or equal to 69. 30. This then becomes our confidence interval for sigma square in the population. And by taking square roots, we get that 5. 18 is less than or equal to sigma, which is less than or equal to 8. 33 with 95 percent probability. So no margin of error computation here. No, directly establish the confidence intervals based on the desired confidence coefficient.

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So, now let us formally define the confidence interval for population variance. Definition time, the confidence interval estimate of population variance is given by sigma square lies between n minus 1 times s square divided by a chi square of alpha by 2 and n minus 1 times s square divided by a chi square of 1 minus alpha by 2, where the chi square values are based on the chi square distribution with n minus 1 degrees of freedom and the 1 minus alpha is our confidence coefficient. Although we did this for a 95 percent confidence interval in our example, the same process can be used to establish an 80 percent confidence interval for a 90 percent confidence interval for Basavaraju's population variance sigma square.

Let us now calculate the 80 percent and the 90 percent confidence intervals for population variance for the same example. Let us start with the 80 percent confidence interval. Therefore, our confidence coefficient is 0. 80, which is 1 minus alpha.

Then, we can directly use the final expression from our definition for the confidence interval of population variance. Sigma square should lie between n minus 1 times s square divided by a chi square of alpha by 2 and n minus 1 times s square divided by the chi square of 1 minus alpha by 2. Since, alpha by 2 is 0. 1, one minus alpha by 2 will be 0. 9.

Let us look at the template for a chi square 35 and find the values of chi square 35 such that the tail probabilities are 0. 9 and 0. 1 respectively. What are these values? For tail probability 0. 9, we get 24. 8 and for tail probability of 0. 1, we get 46. 06. Yes. So, 80 percent of the time, a chi square 35 will take values in between 24. 8 and 46. 06. Then we can substitute these in our expression for confidence interval with s square equal to 40. 7 and n minus 1 equal to 35. And what do we get? 80 percent of the time, the population variance sigma square will lie between 30. 95 and 57. 49. Yes, and taking square roots, we can say that 80 percent of the time, the population standard deviation sigma will lie between 5. 56 and 7. 58.

Tejas and Ashwini, shall we repeat this process to compute the 90 percent confidence intervals? So, in this case, our confidence coefficient is 0. 9, which is 1 minus alpha and we can use directly the final expression for the confidence interval of population variance. Sigma square should lie between n minus 1 times s square divided by a chi square of alpha by 2 and n minus 1 times s square divided by a chi square of 1 minus alpha by 2. And alpha by 2 is 0. 05 in this case. Therefore, 1 minus alpha by 2 is going to be equal to 0. 95.

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Again, we look at the template for a chi square 35 and find the values of chi square 35 such that the tail probabilities are 0. 95 and 0. 05. And what values do we get? For tail probability 0. 95, we get 22. 47 and for tail probability of 0. 05, we get 49. 80. Yes. So, 90 percent of the time, a chi square 35 will take values between 22. 47 and 49. 80. Now, we substitute these values in our expression for the confidence interval with s square equal to 40. 7 and n minus 1 equal to 35. And what do we get? 90 percent of the time the population variance sigma square will lie between 28. 62 and 63. 46. Yes, and if we take the square roots, we can say that 90 percent of the time the population standard deviation will lie between 5. 35 and 7. 97. Because the chi square distribution is not symmetric, We are unable to write this as sample variance plus or minus margin of error, right? Yes, you are right Tejas. This helps us to appreciate the symmetry of the Z and the T distributions a little bit more, doesn't it? Let us do one more example. Let's calculate the confidence interval for Jyothi Hegde's salary data.

Jyothi has salary information from her alumni in her data set. If you recall, the sample size n was equal to 103 in this case. The average salary from the sample was equal to x bar equal to 25. 92 lakhs. The sample standard deviation s was equal to 3. 66 lakhs, which gives us a sample variance of 13. 39 lakhs. And Jyothi wants a 95 percent confidence interval on the salary of her graduates. So the confidence coefficient is 1 minus alpha is equal to 0. 95. And we need to consider a chi square with n minus 1 equals to 102. Yes, you both seem to have this under control. So let's keep going.

For a chi square with 102 degrees of freedom, we want to find the values corresponding to the tail probabilities of 0. 975 and 0. 025. come out to be equal to 75. 95 and 131. 84 respectively. Then sigma square lies between n minus 1 times s square divided by a chi square of alpha by 2 and n minus 1 times s square divided by a chi square of 1 minus alpha by 2. That gives us that 95 percent of the time sigma square will lie between 10. 36 and 17. 98. And that implies that 95 percent of the time the population standard deviation will lie between 3. 22 and 4.24.

Hopefully that has given you some good practice in developing confidence intervals for mean and average proportion and variance of the population. Now we move on to hypothesis testing.

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Hypothesis Testing – Introduction

So, far we have seen how sample data can be used to develop point and interval estimates for the population parameters. We saw how to do this for the population mean mu, the population proportion p and the population variance sigma square. We will continue our discussion on statistical inference and look at another important technique known as hypothesis tests. Hypothesis tests are frequently used to make inferences about the population parameters using sample estimates. The underlying idea behind statistical inference in general and the idea behind establishing confidence intervals and hypothesis tests in particular are very similar.

Essentially, we are trying to take the limited evidence that we have in a sample data and trying to see how best we can use this data to infer information about the parameters of the population from which this sample was drawn. But why care about the population parameters at all? Isn't the sample not enough?

The reason we want to know about the parameters of the population is that more often than the decisions that we make using the sample data are going to impact not just the sample alone, but the larger population from which the sample was drawn. Therefore, before we make those decisions based on the sample data, we want to know as much as we can about the population based on the limited data that we have from the sample.

Being able to have confidence intervals on population parameters like the mean, the proportion and the variance. This helps us understand important aspects of the population. Hypothesis tests will take us further along this path. They will help us test claims about the population parameters like the mean, proportion and variances. Claims like the following, does the sample data provide us with enough evidence to claim that a certain population parameter is greater than a particular value. Is it equal to a particular value of interest? Is it less than a particular value of interest? If these claims about the population parameter are validated by the sample data, then we can confidently make decisions impacting the entire population.

If the guarantees related to these claims are weak, then the evidence from the sample is not strong. And hence, we might be careful and measure. in terms of how we make the decisions that impact an entire population. Now, there is a little bit of structure in terms of how hypothesis

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tests are conducted and some terminology. So, we will spend some time to understand the structure and the terminology and then we will see their applications through several examples.

As with all that we have seen so far and Ashwini, you mentioned this when we started this module also. The actual calculation is fairly simple. But what is more important is to understand the logic behind how to set this up and then complete the calculations.

So let us get started. When we set up a hypothesis test, regardless of whether it is a test for population mean, population variance or any other parameter. There is usually a null hypothesis denoted by H naught and an alternative hypothesis denoted by H a. Setting up the null and alternate hypothesis is important and how they are set up usually depends on the context of the problem and what we are trying to claim from the sample data. Since this is an important step and sometimes tricky. So, we will spend a little bit of time understanding how to set up the null and the alternate hypothesis in different settings.

Next, because a sample is just a sample and it is randomly drawn from the population, probabilities are involved. When we make any claims about a population parameter based on a hypothesis test. This is similar to the probabilities that we had with confidence intervals. Our claims based on a hypothesis test, no matter how carefully done, could be wrong with some probability. And in the context of hypothesis testing, we want to quantify two types of errors that could happen when we make a claim about a population parameter.

These are known as type 1 and type 2 errors. The goal of course is to minimize the amount of both of these types of errors. But as you could imagine, some amount of both types of errors is going to be unavoidable. So as we make these claims, we try to quantify and limit these errors like we did for the confidence intervals. We will look at hypothesis tests related to the population mean, the population proportion and the population variance.

The main purpose of a hypothesis test is that it tries to determine whether a statement about the value of a population parameter should be rejected or should not be rejected. Note that two outcomes of the hypothesis test are, one, whether the statement is rejected or whether the statement should not be rejected. I did not say that the outcomes correspond to whether a statement should be rejected or whether a statement is accepted. There is a subtle difference

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between saying that the statement is accepted based on sample evidence versus a statement is not rejected based on the sample evidence. Seems the same to me.

Can you explain what is the difference? Yes, at first it does look like we are playing with words, but the difference will be clear as we go through some examples. Okay. Thanks. Also, in a hypothesis test, we always have two statements, one known as the null hypothesis and the other known as the alternate hypothesis.

The null hypothesis H naught is commonly referred to as the tentative assumption about a population parameter. The parameter of interest could be the mean mu, the proportion p or variance sigma square. The alternate hypothesis, H a, as it is written, is the opposite of what is stated in the null hypothesis, H naught, corresponding to the population parameter.

The hypothesis testing procedure then uses data from the sample to test which one of these two competing statements, indicated by H naught and H a. The test decides whether the sample provides enough evidence to either reject the null hypothesis or do not reject the null hypothesis.

Let's talk a little bit more about the two outcomes that come out of a hypothesis test. As I said, the decision is always that we either reject the null hypothesis or we do not reject the null hypothesis. When do we reject the null hypothesis? We reject when the sample we used for the test provides definitive evidence that the null hypothesis is not true.

When we reject the null hypothesis, it is proven beyond reasonable doubt that the null hypothesis cannot hold given the sample evidence that we have obtained. On the other hand, if our sample evidence does not provide definitive evidence that the null hypothesis is not true. Then we cannot take the risk of rejecting it.

So the outcome is do not reject the null hypothesis. If the sample casts a reasonable doubt about whether the statement is true or not, then we do not reject the null hypothesis. If we do not reject the null hypothesis, it usually means that the data is inconclusive and it is too risky to make a decision. One way or the other, using a single random sample and the statistics that we

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have obtained from it. In such cases, status quo is usually the decision. To understand the role of the null and alternate hypothesis in hypothesis testing, it is useful to consider a legal analogy.

Let us say that there is a person that is presumed to be guilty of a robbery that has taken place in a store. And a case has been filed in court to try and prove the guilt. So let us say that the store owner files a case. The store then gathers evidence and presents it as evidence in court. And if the evidence is strong and convincing enough, then the person is proclaimed to be guilty of the crime. If the evidence is not convincing enough, the person is presumed to be innocent and let free of any charge.

Now, let us examine what happened in a little bit more detail using the terminology of hypothesis test. In this case, we would say that the null hypothesis is that the person is innocent and the alternate hypothesis is the opposite of the null hypothesis, which is that the person is not innocent.

Realize that in the legal setting, the null hypothesis is always that a person is innocent. That is why we say a person is innocent until proven guilty. Yes, that is right. Now, if the store thinks that they can present enough evidence to prove the guilt, they file a case and then they present evidence to try and prove a claim which is the alternate hypothesis that is the person is not innocent.

So, what the store is trying to prove is written as the alternate hypothesis. Of course, the opposite of this is written as the null hypothesis. Now, during the case, evidence is presented in court. The story may have evidence of the location of the accused during the time of the crime, video footages, fingerprints, and so on. Yes, and if they are able to collect enough evidence. And when they present this evidence, if it is convincing enough, beyond reasonable doubt, that this person has committed the crime, then we have enough evidence to reject the null hypothesis that the person is innocent. And then we have proved the claim that the person is not innocent.

On the other hand, if the evidence was not strong enough, for example, they were unable to precisely determine the location of the accused during the time of the robbery. The camera footage is inconclusive. There are no clear fingerprints that match with the accused. Yes, and

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so on. And then we would say that the evidence collected creates enough doubt, or reasonable doubt, to claim that the person has committed the crime.

And therefore, we say that since there is reasonable doubt, given the available evidence presented, it is too risky to reject the null hypothesis. and claim that the alternate is true. So we do not reject the null hypothesis and the person is assumed to be innocent until proven guilty. Notice that in this case we did not prove the innocence. We were just unable to prove the guilt. There is a difference between proving innocence and not being able to prove the guilt. All that happened was that the court was unable to reject the null hypothesis. because the store was unable to show that the person is not innocent. In fact, in the legal setting, we never prove innocence.

If you are unable to prove the guilt, the person is presumed to be innocent until proven guilty. So there is only two outcomes that could happen either the evidence is strong enough to say that the person is not innocent or the evidence is not strong enough to prove the guilt. And if you are unable to prove the guilt, then we are unable to reject the null hypothesis and the person is assumed to be innocent until proven guilty.

Notice that in this legal case, what the store was trying to prove is the claim that we put in the alternate hypothesis. And if we generalize this to non legal settings, this part of setting up the hypothesis almost usually carries over, in the sense that whatever we are trying to claim is set as the alternate hypothesis at A.

And the opposite statement, of course, is the null hypothesis H naught. And we see whether the sample evidence provides us with enough evidence to reject the null so that we can claim the alternate. The second thing to remember is that the way the hypothesis is set up depends on who is doing the hypothesis test.

So as we look through examples related to hypothesis testing, the first question we ask before we write down the hypothesis is who is trying to make the claim about a population parameter? And the second question is what is the specific claim that they are trying to make regarding a population parameter using the sample data?

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Once you have answers to these two questions, then we can set up the null hypothesis and the alternate hypothesis and carry out the calculations for the test. Therefore, it is important to structure the hypothesis appropriately. So, that the test conclusion provides a necessary information to the person conducting the text.

And we will see that the context of the situation is important to decide how the hypothesis, especially the null hypothesis and the alternate hypothesis should be stated. This usually requires a little bit of practice to understand how the hypothesis should be stated in a given context. But as I mentioned. Two questions are usually helpful in setting the hypothesis correctly.

The first is to ask who is conducting the test when trying to make a claim about the population parameter. And the second is to ask what is the person trying to claim specifically about the population parameter when they are trying to use sample data that has been collected and do the test.

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Null & Alternative Hypothesis

In most cases, hypothesis tests take one of the following three forms and usually we write one of the following three types of hypothesis tests. An upper tail test, a lower tail test or a two tail test. Let us consider what these three forms are using the context of a hypothesis test about the population parameter mu, the mean of the population.

Using the data from the sample, we may want to claim one of the following. One, the population mean mu is less than a claimed value A or two, a population mean mu is more than a claimed value A or three, the population mean mu is not equal to a certain claimed value A. If you are trying to use the data to prove a claim that the population mean mu is less than a claimed value a, then that becomes the alternate hypothesis.

The opposite of the statement that the population mean mu is greater or equal to a, then becomes the null hypothesis. Since, the statement in the alternate hypothesis is the less than type mu less than a. It is called a lower tail test. The specific reason is related to how something called the p value is computed, and that will become clearer a little bit later.

If you are trying to use the data to prove a claim that the population mean mu is greater than a claimed value, then that becomes the alternate hypothesis. The opposite of the statement that the population mean mu. is less than or equal to a becomes the null hypothesis. And since the statement in the alternate hypothesis is of the greater than type, that is mu greater than a, it is called an upper tail test.

Finally, if you are trying to use the data to prove a claim that the population mean mu is not equal to a claimed value a, then that becomes the alternate hypothesis. The opposite of the statement that the population mean mu is exactly equal to A, then becomes the null hypothesis. Since the statement in the alternate hypothesis is of the not equal to type, that is mu not equal to A, this is called as a two tailed test. Notice that by convention, the equality sign is always in the null hypothesis. Quite a bit of terminology. Yes, but with some examples it will become clear.

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Let us look at some examples to see how to set up the null and alternate hypothesis before we go any further. Santosh Mishra shops regularly at the Jalan Supermarket in Durgakund Road in Varanasi. He likes the variety of products that they have and the fact that they are fresh and reasonably priced. However, recently he has had an issue while buying his regular supplies of lentils at the store. He has found that although the packet is labeled 500 grams, the amount of lentils in the packet is less than the labeled quantity. At first he dismissed it, but then when it started to happen frequently, He suspected that this is a deliberate effort by the store. His friends and neighbors have had similar experiences. So they filed a complaint with the Consumer Affairs Department to investigate whether Jalan's supermarket chain is deliberately underfilling lentils in a packet.

The Consumer Affairs Department or CAD does a study in response to this complaint. And they collect a random sample of 50 lentil packets from various stores of the Jalan supermarket chain. The sample yields a sample mean of 497 grams with a sample standard deviation of 7 grams. How should consumer affairs department structure a hypothesis test that would determine whether the supermarket is deliberately underfilling the lentil packets to a quantity less than 500 grams?

And if they did the hypothesis test, what would be the conclusion for a significance level of 4%? As mentioned earlier, a hypothesis test has two statements, the null hypothesis and the alternate hypothesis, and one is the opposite of the other. In this case, we have an issue related to the weight of lentils in a packet labeled 500 grams.

The suspicion is that the amount of lentils is deliberately being set at less than 500 grams. The population in this case is a set of all 500 grams labeled lentil packets packaged by Jalan Supermarket. The population parameter for which the claim is being tested is the population average mu, the amount of lentil in a packet labeled 500 grams.

Now, When any packaging process fills lentils in a packet label 500 grams, you are unlikely to get exactly 500 grams. Sometimes it is going to be less and sometimes it is going to be more. And the sample of 50 packets indicates that a sample average of X bar of 497 and a sample standard deviation S of 7 grams.

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Indeed, the question now is whether this sample provides enough evidence to claim that Jalan supermarket is deliberately underfilling the packets or whether the variation that the sample indicates is due to the various sources of randomness that one can expect when the average quantity in the packets is actually 500 grams.

While setting up the null and alternate hypothesis, usually the claim being tested using the sample data is written as the alternate hypothesis. To decide the exact form of the alternate hypothesis, it is useful to think of it from two perspectives. First, who is the one that is trying to make a claim about the population parameter?

And two, what is the specific claim that they are trying to show? In this case, it is the consumers affairs department or CAD that is trying to use the sample data to stake a claim that Jalan Supermarket is derivatively underfilling their 500 grams lentil packets. And what is the Consumer Affairs Department trying to show?

They are trying to show that the average weight of packets in the population mu is less than 500 grams. And so therefore, mu is less than 500 grams would be the alternate hypothesis. What would be the null hypothesis then? The null hypothesis then would be the opposite of this statement, which would be that the average amount of lentils in a packet in the population, mu, is greater or equal to 500 grams.

If they conduct tests, what could the outcomes be? If the sample data provides enough evidence, reasonable doubt to reject the null hypothesis that says that mu is greater or equal to 500 grams, then the consumer affairs department has enough evidence to claim that mu is less than 500 grams and that Jalans is deliberately underfilling.

Alternatively, if the sample data does not have enough evidence to be conclusive, there is reasonable doubt due to the randomness. Then, Consumer Affairs Department does not have enough evidence to reject the null hypothesis that mu is greater or equal to 500 grams. So, it cannot claim that mu is less than 500 grams.

The hypothesis test is inconclusive and status quo will be accepted. And Jalan's supermarket will be deemed to be innocent. Yes. Note that not rejecting the null hypothesis does not prove

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their innocence. The sample just does not prove that Jalans is filling 500 grams. It is just that we do not have enough evidence to prove that they are underfilling. Does that make sense? Yes, but can we do another one? Sure.

Geeta Kumari is the manager at the Samsung Service Center on Bypass Road, Bareilly. They have a policy that all service requests will be responded to with a mean time of 48 hours or less. As a manager of the service center, Geeta periodically checks to see whether their service requests are actually adhering to the claim that they make to their customers.

She wants to make use of a sample of service work orders to determine whether the mean response time has increased to more than 48 hours. Suppose she selects a random sample of 40 service work orders and the sample indicates a mean of 52 hours and a sample standard deviation of 10 hours, should Geeta Kumari be worried that the average time for responding to a service request has actually exceeded to 48 hours that they claim. How should she structure hypothesis test to determine whether the mean response time for service work order is indeed more than 48 hours. If you conducted this hypothesis test what would the conclusion be for a significance level of 3%. In this case Geetha Kumari is worried whether the time to respond to a service requests is more than 48 hours. The population in this case is a set of all service work orders handled by the service center.

The population parameter for which a claim is being tested is the population average, mu, the average time to respond to a service work order. Now, of course, all service work orders will not take exactly 48 hours or less. Some would take more than 48 hours. Sometimes it is going to be less, sometimes it is going to be more.

And the sample average of 40 work orders indicates that the sample average X bar is 52 hours and the sample standard deviation S is 10 hours. So the question is whether the sample provides enough evidence to claim that the average response time for work orders in the population is sufficient not just the ones in the sample exceeds 48 hours or whether the variation that the sample is showing is just due to the various sources of randomness that one can expect when the response time is indeed less than 48 hours on an average.

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While setting up the null and the alternate hypothesis, usually the claim being tested using the sample data is written as the alternate hypothesis. Yes, Tejas. And as I said earlier, to decide the exact form of the alternate hypothesis, it is useful to think of it from two perspectives. First, who is the one that is trying to make a claim about the population parameter?

And what is the specific claim that they are trying to show? In this case, Geeta Kumari is trying to use the sample data to test the claim. Yes, that's true. And what is she trying to show? She is trying to verify if the average response time has increased to more than 48 hours. Yes, so therefore, mu greater than 48 hours would be the alternate hypothesis.

The null hypothesis then would be the opposite of this statement, which would be that the average time to respond to a work order, the population mu, is less than or equal to 48 hours. If they conduct tests, what could the outcome be? If the sample data provides enough evidence beyond reasonable doubt to reject the null hypothesis that says that mu is less than or equal to 48 hours, then Geeta Kumari should take corrective action, since she has enough evidence to claim that mu is greater than 48 hours.

Alternatively, if the sample data does not have enough evidence to be conclusive, then there is reasonable doubt due to the randomness. And Geeta Kumari would not have enough evidence to reject the null hypothesis that mu is less than or equal to 48 hours. So she cannot claim that mu greater than 48 hours.

Yes, and so the hypothesis test would be inconclusive and status quo will be accepted. Again, note that not rejecting the null hypothesis does not prove the null hypothesis. It is just that the sample does not prove that the response time is is more than 48 hours. Is that okay? One more? Let's take another instance.

Ibrahim Khan runs a chain of petrol stations in the Agra to Mathura NH44 corridor. He takes great pride in the service that he provides to his customers. The filling stations are calibrated to ensure that the right amount of petrol is dispensed to his customers. However, recently he's been having some doubts about the accuracy of his equipment.

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He is wondering whether when customers are asking for 30 liters of petrol to be filled in their vehicle, whether the equipment is actually filling 30 liters or not. If the equipment is not dispensing the right amount of petrol, then the equipment has to be temporarily discontinued and corrective maintenance has to be taken.

To verify, Ibrahim selects a random sample of 32 transactions and he compares the meter reading of 30 liters against the actual amount of petrol dispensed. He finds that the average amount of petrol dispensed in these 32 samples is 30. 3 liters. The standard deviation in the sample is found to be 0. 5 liters. He is wondering how to interpret the sample data. Should Ibrahim be worried? How should Ibrahim set up a hypothesis test to determine, should he take corrective actions and try to recalibrate the pumping equipment? What would be the outcome of the hypothesis test if it was conducted at a significance level of 5%? In this case, Ibrahim is worried whether the amount of petrol dispensed is 30 liters or not.

The population in this case is the set of transactions where a request was to fill 30 liters of petrol. The population parameter for which the claim is being tested is the population average mu. The average amount of petrol dispensed when 30 liters was requested. Now, of course, all transactions will not exactly dispense 30 liters.

Sometimes it's going to be less, sometimes it's going to be slightly more. And the sample of 32 transactions indicates a sample average x bar of 30. 3 liters and a standard deviation of 0. 5 liters. So the question is whether the sample provides enough evidence to claim that the average amount of petrol dispensed across all transactions in the population, not just the ones in the sample, has an average of other than 30 liters, or whether the variation that the sample is showing is just due to the various sources of randomness that one could expect when the amount is dispensed is 30 liters on an average.

We know that usually the claim being tested using the sample data is written as the alternate hypothesis. Again, as was mentioned earlier, to decide on the exact form of the alternate hypothesis, it is useful to think of it from two perspectives. First, who is the one that is trying to make a claim about the population parameter?

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And what is the claim that they are trying to show? Yes, and in this case, Ibrahim Khan is trying to use the sample data to test a claim. And what is he trying to show? He is trying to verify if the average amount of petrol dispensed is not equals to 30 litres. Yes, he should be worried about dispensing more and dispensing less than 30 litres.

If the equipment is faulty and it is dispensing more than 30 litres on an average, he is giving petrol for free and that is not good for his business. If the equipment is dispensing less than 30 litres on average, He could be accused of manipulating the equipment to deliberately underfill and cheat his customers.

Therefore, mu not equal to 30 liters would be the alternate hypothesis. That makes sense for Ibrahim. The null hypothesis then would be the opposite of the statement, which would be that the average amount dispensed, the population mean mu is equal to 30 liters. And if he conducts the hypothesis test, What could the outcomes be?

If the sample data provides enough evidence beyond reasonable doubt to reject the null hypothesis that says that mu equal to 30 liters, then Ibrahim would have to take corrective action and recalibrate the pumping equipment, since he has enough evidence to claim that mu is not equal to 30 liters.

Alternatively, if the sample data does not have enough evidence to be conclusive, there is reasonable doubt due to the randomness. Then, Ibrahim Khan does not have enough evidence to reject the null hypothesis that mu is equal to 30 liters. So, he need not worry that mu is not equal to 30 liters. The hypothesis test is inconclusive and status quo will be accepted.

Again, note that not rejecting the null hypothesis does not prove the null hypothesis. It is just that the sample does not prove that mu is not equal to 30 liters. These examples definitely helped.

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Null & Alternative Hypothesis

The null and alternate hypothesis are competing statements about the population. Therefore, either the null hypothesis is true or the alternate hypothesis is true, but not both and the hypothesis testing procedure should reject the null hypothesis when the null hypothesis is not true and it should not reject the null hypothesis when the null hypothesis is true.

Unfortunately, correct conclusions are not always possible because hypothesis tests use sample data and therefore, there is always a possibility of error during hypothesis tests. There are two kinds of errors that can be made, a type 1 error and a type 2 error. I've heard of this term. I've heard these terms, but never knew what they were.

Yes. So let us see them both. Let us say that the conclusion of a hypothesis test based on the sample was to reject the null hypothesis. Now, if the sample indeed came from a population where the null hypothesis is true, and we ended up rejecting the null hypothesis, then we wrongly rejected the null hypothesis.

And we say that we have made a type 1 error. So a type 1 error results in wrongly reject the null hypothesis when the null hypothesis is true. Yes, that is right. However, if the sample came from a population where the alternate is true, then rejecting the null hypothesis based on the sample information is the correct decision.

Next, consider what can happen if the conclusion of the hypothesis test is to do not reject the null hypothesis based on the sample information. If the sample came from a population where the null hypothesis is indeed true, then this conclusion of the hypothesis test is correct. However, if the sample came from a population where the alternate hypothesis is true, then we should have rejected the null hypothesis and we did not. This corresponds to a type 2 error. That is we do not reject the null hypothesis when in fact the alternative is true. Yes, that is correct. Now let us see these errors in the context of one of the examples that we just discussed. Recall Geeta Kumari is worried about the response time of her service calls and whether they exceeded 48 hours.

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We had set up the null hypothesis to be mu less than or equal to 48 hours. and the alternate to be mu greater than 48 hours. The alternate hypothesis of mu greater than 48 hours suggests that Geeta is trying to look for sample evidence to support the conclusion that the average response time to respond to service calls in the population is greater than 48 hours.

In this case, a type 1 error wrongly rejecting the null hypothesis when it is true corresponds to claiming based on the sample evidence that the average response time has indeed exceeded 48 hours, when in fact the average response time was 48 hours or less. In contrast, a type 2 error of wrongly accepting the null hypothesis when it is not true corresponds to Geeta concluding based on the sample evidence that the average response time is less than or equal to 48 hours, when in fact the average response time is more than 48 hours.

In Geeta Kumari's example, the null hypothesis is mu less than or equal to 48. Suppose the null hypothesis is true at equality, that is mu is exactly equal to 48. Then the probability of making a type one error when the null hypothesis is true at equality is called as the level of significance and is denoted by the Greek letter alpha.

In Geeta's case, the level of significance is the probability of rejecting the null hypothesis of mu less than or equal to 48 when mu is exactly equal to 48. The most common choices for alpha are 0. 05 and 0. 1. Usually, the person conducting a hypothesis test chooses the level of significance. And by choosing the alpha desire, they control the probability of that type one error.

If the cost of making a type one error is high, small values of alpha are preferred. If the cost of making a type one error is not too high, then larger values of alpha can be accepted. Hypothesis tests that control only for type one error are known as significant tests, and although in most cases we mainly control for the type one error controlling for the type one error is not necessarily control the probability of making a type 2 error. So, usually there is a balance that needs to be struck in terms of controlling for type 1 and type 2 errors while conducting a hypothesis test. Yes, that's right. Now, let us see how to conduct these tests.

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Hypothesis Testing for Population Mean

The process for conducting a hypothesis test usually has five main steps. The first step is to identify the null hypothesis and the alternate hypothesis. The second step is to decide on the significance level, alpha, the maximum allowable type one error. The third step in the process is to start with the assumption that the null hypothesis is true at equality.

Then, based on this assumption and the sample data collected, we compute the value of the test statistic. In step 4, there are two possible options. The first option is to use the p value approach. The second option is to use the critical value approach. Let us talk about the p value approach. In the p value approach, based on the hypothesis identified, the level of significance and the test statistic computed, we calculate the probability of making a type 1 error, assuming that the null hypothesis is true at equality.

If this p value is smaller than the level of significance alpha, the maximum allowable type 1 error, then we reject the null hypothesis. If the p value is more than the level of significance, then we do not reject the null hypothesis. If we choose to use the critical value approach in step 4 instead of the p value approach, we do the following.

We use the significance level alpha to compute the critical value and a rejection rule based on the hypothesis. And based on whether the hypothesis is upper tail, lower tail or two tail test, we will use the value of the test statistic and apply the rejection rule and make our decision on the null hypothesis.

That is whether to reject it or to not reject it. Regardless of which approach we take in step 4, the p value or the critical value, our decision on the hypothesis will be the same. And the final step is step 5, where we provide the managerial implication of our decision. What does rejecting the null hypothesis mean for the decision maker?

And what does deciding not to reject the null hypothesis imply for decision maker. Let us see the application using the examples that we just discussed. Let us start with the Jalan Supermarket example. Let us start with Santosh Mishra's complaint to the Consumer Affairs

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Department and their response to this complaint. They collected a random sample of 50 lentil packets from the various stores of Jalan Supermarket chain.

The sample gave a sample mean of 497 grams and a sample standard deviation of 7 grams. We decided that the null hypothesis in this case would correspond to a population mean mu greater or equal to 500 grams and what was the alternate? Then the alternate hypothesis, the claim Consumer Affairs Department is trying to prove through sample data is population mean mu less than 500 grams.

Yes. And suppose the Consumer Affairs Department did the hypothesis test as set up, we want to know what would be their conclusion for a level of significance of 4%. Having set up the null and the alternate hypothesis, next we assume that the null hypothesis is true at equality. That is we assume that we are indeed sampling from a population where the population mean is equal to 500 grams.

Then, what do we know about the distribution of the sample average X bar? Since the sample size n is 50, which is greater than 30, the distribution of X bar is approximately normal with standard error sigma over root n. Yes, and since we don't know sigma, we compute the test statistic to be equal to X bar minus mu divided by S over root n.

Now, in general, X bar minus mu divided by s over root n is a random variable. And since the distribution of X bar is normal, the distribution of X bar minus mu divided by sigma over root n is the standard normal. And therefore, the distribution of X bar minus mu divided by s over root n would be a t distribution with n minus 1 or 49 degrees of freedom in this case.

And what our sample has provided is one random instance of this test statistic coming from a t distribution with 49 degrees of freedom. What is the specific value that our sample has given us? We have s over root n equal to be 8 divided by root 50 which is 1. 13. Yes, and since we have assumed the hypothesis to be true at equality, our test statistic would come out to be equal to 497 minus 500 divided by the 1.13 and that comes out to be equal to minus 2. 65. The next step is to compute the p value which is the probability of a type 1 error based on this test statistic. The probability of a type 1 error would be equal to the probability of getting a test statistic that is minus 2. 65 or less when the null hypothesis is true.

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So, we want the probability of a t 49 taking a value of minus 2. 65 or less. We compute this value from the t distribution template. I think the p value comes out to be 0. 005. Yes, that is correct. And since the p value is less than alpha of 0. 04. Therefore, we reject the null hypothesis and we conclude that there is enough evidence in the sample to claim that the supermarket must be deliberately underfilling the lentil packets labeled as 500 grams.

If the null hypothesis were true and the population indeed had packets with 500 grams or more, it is highly unlikely that we should be getting a sample average of X bar of 497 based on a random sample of 50 packets from that population. But since our sample average is indeed as low as 497, it is very highly unlikely that this came from a population where the null hypothesis is true.

And so we could say that there is enough evidence to reject the null hypothesis and claim that it comes from a population where the average weight mu is less than 500 grams. Okay. Now, let us test the hypothesis for Geeta Kumari. As you recall, Geeta Kumari is a manager at the Samsung service center on bypass road in Bareilly.

She wants to use the sample of service work orders to determine whether the mean response time for her service request has increased to more than 48 hours. So, she selects a random sample of 40 service work orders. And the sample indicates a mean of 52 hours and a sample standard deviation of 10 hours.

We decided that the null hypothesis in this case would correspond to a population mean mu less than or equal to 48 hours. And what is the alternate here? Then the alternate hypothesis, the claim Gita is trying to prove through sample data is population mean mu more than 48 hours. Correct. And so now, we want to know what is the conclusion if we did this test for a significance level of 3 percent. Having set up the null and the alternate hypothesis, next we assume as we did earlier that the null hypothesis is true at equality. That is, we assume that we are sampling from a population where the population mean mu is indeed equal to 48 hours.

And since the sample size n is 40, which is greater than 30, the distribution of x bar will be approximately normal with standard error sigma over root n. Again, since we don't know sigma, we compute the test statistic to be x bar minus mu divided by s over root n. And the distribution

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of x bar minus mu divided by s over root n would be a t distribution with n minus 1 or 39 degrees of freedom.

And what our sample has provided is one random instance of these test statistics coming from a t distribution with 39 degrees of freedom. Yes, and what is the specific value that our sample has given us? We have s over root n equal to be 10 divided by root 40 which is 1. 58. Yes, and since we have assumed the hypothesis to be true at equality, our test statistic would come out to be equal to 52 minus 48 divided by the 1.

58 that you just got and that comes out to be equal to 2. 53. So now, the next step is to compute the p value, which is the probability of a type 1 error based on this test statistic. The probability of a type 1 error would be equal to the probability of getting a test statistic of 2. 53 or more when the null hypothesis is true.

We want the probability of a t 39 taking a value of 2. 53 or more. We compute this value from the t distribution template. The p value comes out to be 0. 008. And since the p value is less than the alpha of 0. 03, we reject the null hypothesis. And we can conclude that there is enough evidence in the sample to claim that the response times are on an average more than 48 hours in the population.

If the null hypothesis were true and the population indeed had a response time of 48 hours or less, it is highly unlikely that we should be getting a sample average of X bar of 52 hours based on a random sample of 48 work orders from that population. But, since our sample average response time is indeed as high as 52 hours, it is very highly unlikely that this actually came from a population where the null hypothesis is true.

And so, we could say that there is enough evidence to reject the null hypothesis and claim that it comes from a population where the average response times mu is more than 48 hours. Now, what if Geeta selects a random sample of 40 service work orders and the sample indicates a mean of 50 hours and a sample standard deviation of 10 hours.

The null hypothesis would correspond to the population mean mu less than or equal to 48 and the alternate hypothesis, the claim Geeta is trying to prove through sample data is that the

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population mean mu is more than 48 hours. Now, for this revised data, let us say we want to know what is the conclusion of a hypothesis test for a significance level of 3 percent.

The setup of the null and the alternate hypothesis still remains the same. The only change is in the computation of the test statistic. Since we have assumed hypothesis to be true at equality, this would come out to be equal to 50 minus 48 divided by 1. 58, which comes out to be equal to 1. 26. Yes, and the next step would be to compute the p value, which is the probability of a type 1 error based on this test statistic.

We want the probability of a T 39 taking a value of 1. 26 or more. We compute this value from the T distribution template. The p value comes out to be 0. 107 and since the p value is more than the alpha of 0. 107. And since the p value is 0. 107 and is more than the alpha value. We do not reject the null hypothesis and we conclude that there is not enough evidence in the sample to claim that the response times are on an average more than 48 hours in the population.

Now, let us go and help Ibrahim in Agra. If you recall, Ibrahim Khan runs a chain of petrol stations in the Agra to Mathura NH44 corridor. He is wondering whether when his customers are asking for 30 liters of petrol to be filled in their vehicle, whether the equipment is indeed filling 30 liters. To verify, Ibrahim selects a random sample of 32 transactions and he compares the meter reading of 30 liters against the actual amount of petrol dispensed.

He finds that the average amount of petrol dispensed in these samples is 30. 3 liters. The standard deviation of the sample is found to be equal to 0. 5 liters. How should Ibrahim set up a hypothesis test to determine if he should take corrective actions and try to recalibrate the pumping equipment? We decided that Ibrahim will be concerned if the equipment is filling more than 30 liters and less than 30 liters.

Therefore, mu not equal to 30 liters would be the alternate hypothesis. The null hypothesis then would be that the population mu is equal to 30 liters. Yes, and we also decided to conduct the hypothesis test at a significance level of 5 percent. Having set up the null and alternate hypothesis, next we assume that the null hypothesis is true at equality.

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That is, we assume that we are indeed sampling from a population where the population mean mu is equal to 30 liters. And since our sample size n is 32 which is greater than 30, the distribution of x bar is approximately normal with the standard error of sigma over root n. Again since we do not know sigma, we compute the test statistic to be x bar minus mu divided by s over root n.

And the distribution of x bar minus mu divided by s over root n in this case would be a t distribution with n minus 1 or 31 degrees of freedom. And what our sample has provided is one random instance of this test statistic coming from a t distribution with 31 degrees of freedom. And what is the specific value that our sample has given us?

We have s over root n equal to be 0. 5 divided by root 32 which is 0. 09. Yes, and since we have assumed that the hypothesis to be true at equality, the test statistic would be coming out to be equal to 30. 3 minus 30 divided by the 0. 09 and that comes out to be equal to 3. 39. The next step then is to compute the P value, which is the probability of type 1 error based on this test statistic.

The probability of type 1 error would be equal to the probability of getting a test statistic of 3. 39 or more plus the probability of getting a test statistic of minus 3. 39 or less when the null hypothesis is true. Due to the symmetry, this would be two times the probability that a T 31 takes a value of 3.

39 or more. We compute this value from the T distribution template. Is the p value 0. 00? Yes, it is. And since the p value is less than the alpha of 0. 05 or 5 percent, we reject the null hypothesis. And we conclude that there is enough evidence in the sample to claim that the equipment is not dispensing the set quantity of 30 liters.

If the null hypothesis were true, it is highly unlikely that we should be getting a sample average of X bar of 30. 3 based on a random sample of 32 transactions from that population. But since the X bar is indeed 30. 3. It is highly unlikely that it came from a population where the null hypothesis was true. And therefore, we say that there is enough evidence to reject the null hypothesis.

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Summary of Hypothesis Testing for Population Mean

So, Tejas and Ashwini, we saw examples of all three types of hypothesis, a lower tail test corresponding to the lentil example, an upper tail test corresponding to the service work order example, and a two tail test corresponding to the petrol station example. Did those examples help you understand how to conduct a hypothesis test?

Yes, they all had the same general process, but I am bit confused about how you computed the p value in each case. Can you summarize them again? Sure. Let me try and recap. Remember, the claim being tested is usually in the alternate hypothesis. And a lower tail test is a test where the alternate hypothesis has a less than form.

That is, if you are testing for the population mean, then in a lower tail test, the alternate hypothesis is of the form mu less than a value, say A. The null hypothesis would then be mu greater than or equal to A. We compute the test statistic assuming that the null hypothesis is true at equality. The test statistics would be a t n minus 1 random variable.

The p value is then computed as the probability that a t n minus 1 takes a value less than the value of the test statistic that we computed. For lower tail test, the p value is probability which is less than the test statistic. Yes, that is right. And you reject if the p value is less than or equal to our significance level alpha.

For an upper tail test, the alternate hypothesis has a greater than form. That is, if you are testing for the population mean, then in an upper tail test, the alternate hypothesis is of the form mu greater than a value, say A. The null hypothesis would then be mu less than or equal to A. We compute the test statistic, assuming the null hypothesis is true at equality.

The test statistic again would be a T n minus 1 random variable. The p value is computed as the probability that a T n minus 1 takes a value greater than the value of the test statistic that we computed. Okay, so for upper tail test, the p value is probability, it is greater than the test statistic. Yes, and for a two tailed test, the alternate hypothesis has a not equal to form.

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That is, if you are testing for the population mean, then in a two tailed test, the alternate hypothesis is of the form mu not equal to a value, say A. The null hypothesis would then be mu equal to A. We compute the test statistic, assuming the null hypothesis is true at equality. The test statistic again would be a t n minus 1 random variable.

And if the test statistic is positive, then we compute the probability that a t n minus 1 takes a value greater than the test statistic. And the p value is double of this probability because of the symmetry. If the test statistic is negative, then we compute the probability that a t n minus 1 takes a value less than the test statistic.

Again, the p value is double this probability because of symmetry. And in all types of tests, we reject the null hypothesis if the p value is less than or equal to the significance level. We reject the null hypothesis even if the p value is exactly equal to alpha. Yes, we reject for all values less than and equal to alpha.

So, to summarize, hypothesis test for population parameter mu can be of three types, a lower tail test, an upper tail test, and a two tail test. In each case, we write the null hypothesis and the alternate hypothesis. Once you have written the statements of the null and the alternate hypothesis, then you compute the test statistic using the sample data, assuming that the null hypothesis is true at equality.

The test statistic is a t random variable with n minus 1 degrees of freedom, where n is the sample size. We compute the specific value of this test statistic using the sample data. Then depending upon the nature of the test, lower tail, upper tail or two tail, you compute the p value as we just discussed.

If the p value is less than or equal to alpha, then you reject the null hypothesis. Otherwise, you do not reject the null hypothesis. Now a special case of this is the case where you have information about the sigma of the population and if that is the case, then the test statistic becomes a Z random variable, the standard normal.

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And instead of using the t distribution, you can use the standard normal distribution to compute the probabilities. The rest of the process remains the same. Does that help, Ashwini and Tejas? Yes, definitely. Thanks a lot. You are welcome.

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Hypothesis Testing of Population Proportion

Next, we are going to look at hypothesis test for the population proportion p. We already saw how to write confidence intervals for the population parameter p using the sample estimate p bar. Yes, the process was very similar to the way we wrote confidence intervals for the population mean mu. Yes, and similarly, the hypothesis test for the population parameter p will be similar to the hypothesis test that we just saw for the population mean mu. Like we had for the mean mu, the hypothesis test for the population parameter p will also have a null and alternate hypothesis. Here too the alternate hypothesis would be written based on the claim that we are trying to make regarding the population parameter p.

Like the hypothesis test for population mean mu, we would have a lower tail test, an upper tail test and a two tail test. For the population parameter p, for the lower tail test, the alternate hypothesis has a less than form. That is, the alternate hypothesis is of the form p less than a certain value, say p 0.

So, the null hypothesis would then be p greater than or equal to p 0. Oh, just mu replaced by p, since it is for population proportion now. Yes, exactly. And for an upper tail test, the alternate hypothesis has a greater than form. That is, the alternate hypothesis is of the form p greater than a value, say p 0. The null hypothesis would then be p less than or equal to p 0.

And for a two tailed test, the alternate hypothesis has a not equal to form. That is, the alternate hypothesis is of the form p not equal to a value, say p 0. The null hypothesis would then be equal to p equal to p 0. And the equality is always in the null hypothesis. Yes, that is right. What about the steps to test the hypothesis?

Yes, they are very similar to what we used for the mean. Write the null and the alternate hypothesis and decide on the significance level alpha. Then, you compute the test statistic, compute the p value and reject the null hypothesis if the p value is less than null. Is the test statistic T random variable?

No, actually let us think through what it should be. So, what do we know about the sampling distribution of p bar? If n is large enough, the sampling distribution of p bar is approximately

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normal. Yes, and centered at the population parameter p with a standard deviation sigma of p bar given by the square root of p times 1 minus p divided by n.

So, assuming the null hypothesis is true at equality. We will compute the value of the test statistic to be equal to p bar minus p 0 divided by the square root of p 0 times 1 minus p 0 divided by n. Now, what type of random variable would this be? We know that p bar is approximately normal and since we are assuming the null hypothesis is true at equality, we are assuming that p bar comes from a sample

centered at a population proportion equal to p zero. Then if we subtract the mean p zero from p bar and divide by its standard deviation, what do we get? We get z random variable. Yes, Ashwini. And so the test statistic here is a standard normal random variable. Then we follow the process assuming that this is a z random variable rather than a random variable.

Yes, that's right. And the rest of the process. Shall we see some examples?

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Example of Hypothesis of Population Proportion

Anil Verma of Swiggy is working out of his Prayagraj office. He is considering a statewide promotion that involves mailing discount coupons to all his customers. This promotion will be considered as a success if more than 10 percent of all of those receiving coupons use them. Before going statewide with the promotion

coupons were sent to a random sample of 80 customers. He found that 12 out of these 80 customers use the coupons. He wants us to obtain a point estimate of the proportion of customers that will use the coupons. He wants us to construct a hypothesis test to determine if the sample study provides enough evidence for him to claim that more than 10 percent of the coupon recipients will use them.

And if the hypothesis test is done using a significance level of 5%, what would be the outcome? Ashwini and Tejas, how would you set up the null and the alternate hypothesis? Who is trying to make the claim using sample data and what are they trying to claim? Anil is trying to use sample data to see if there is evidence that more than 10 percent of coupon recipients are using them.

Yes, that is right. So what is the claim that we put as the alternate hypothesis? The alternate is going to be population proportion p greater than p0 where p0 is 10%. And so the null hypothesis will be the opposite statement which is that the population proportion p is less than or equal to p 0 where p 0 is 10 percent.

So this is an upper tail test. Yes, that is right. Now, Anil has asked us to use a significance level of 5 percent for the test. Now, what is the evidence from the sample? We have 12 successes out of n equals to 80. Therefore, the sample proportion p bar is 12 divided by 80, which is 15 percent. And that's our point estimate.

And the question is, does this 15 percent in the sample provide enough evidence that the proportion is more than 10 percent in the population? Next, we assume that the null hypothesis is true at equality. That is, we are indeed sampling from a population where p is less than or equal to 10 percent. Now, compute the test statistic.

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We first need the standard error sigma of p bar. Since np and nq are both more than 5, we can assume that p bar comes from a normal distribution with mean equal to the claim in the null hypothesis, which is 0. 1 in this case, and the standard error equal to the square root of p 0 times 1 minus p 0 divided by n.

Can you calculate that? It is the square root of 0. 1 multiplied by 0. 9 divided by 80, which is 0. 03. Yes. So, that means our test statistic P bar minus P 0 divided by sigma P bar is a standard normal random variable. And the particular value that we have for this statistic based on the random sample is equal to 0. 15 minus 0. 1 divided by 0. 03 which comes out to be 1. 49. Now, what is the P value in an upper tail test? For mean, we said it was the tail probability corresponding to the test statistic. Is it the same here for proportion? Yes, same logic.

So we want the probability that a standard normal random variable takes a value greater than 1. 49. From the standard normal template, we find that the tail probability to be 0. 068. Yes, so the p value is 0. 068 which is more than our significance level of 5%. So we cannot reject the null hypothesis. Yes, that's true. The sample does not provide enough evidence to claim that the proportion of people using coupons in the population is more than 10%.

Interesting, isn't it? Even though the sample has a 15 percent usage, why do you think this does not carry forward into the population? Is it because of the standard error? The 0. 03 we got is relatively high and not allowing us to make claims on the population. Yes, that is exactly correct. We have to be careful in making claims and decisions based on limited data.

Now I see the values of doing these tests while making decisions. Yes. So now let's do another example. We are going to Jhansi. Pushpa Yadav is the chief engineer working at Jhansi. Investigating a new design for sleeper trains. She has a study that claims that 52 percent of the train passengers do not feel the sleeper arrangements are adequate for long distance express trains.

She does a random sample of 510 individuals traveling long distance by Indian railways and she finds that 280 indicated that sleeper arrangements were adequate on the long distance express trains. She wants to determine if less than 50 percent travelers on the long distance trains feel that sleeper arrangements are inadequate.

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She would like us to conduct a hypothesis test using a significance level of 3%. How do we set this up and what can we conclude based on the sample data? Ashwini. Do you want to set up the null and the alternate hypothesis? Who is trying to make a claim using the sample data and what are they trying to claim?

Pushpa is trying to use sample data to see if there is evidence that less than 50 percent of travellers feels that sleeper arrangements are inadequate. Yes, that is right. So what's the claim that we put as the alternate hypothesis? The alternate is going to be population proportion p less than p 0, where p 0 is 0. 5. Yes, and so the null hypothesis will be the opposite statement, which is that the population proportion p is greater than or equal to p 0, where p 0 is 0. 5. And that would make this a lower tail test. Now, Pushpa has asked us to use a significance level of 3 percent for the test. What is the evidence from the sample?

We have 280 successes out of n equals to 510. Therefore, the sample proportion p bar is 280 divided by 510, which is 0. 55. Okay. By the way, what is success here based on our hypothesis? The number that feel arrangements are inadequate. Yes, and what does the 280 represent? The number that feel arrangements are adequate.

Oh, so P bar should be 510 minus 280 divided by 510, which is 0. 45. Yes, so watch the statements carefully. Okay, next, we assume that the null hypothesis is true at equality. That is, we are indeed sampling from a population where p is greater or equal to 0. 5. Now, compute the test statistic. For that, we need the standard error sigma p bar.

Since np and nq are both more than 5, we can assume that p bar comes from a normal distribution with mean equal to claim in the null hypothesis, which is 0. 5 and a standard error equal to 0 times 1 minus p 0 divided by n. Can you calculate that? It is the square root of 0. 5 multiplied by 0. 5 divided by 510, which is 0. 022. Yes, and so that means our test statistic p bar minus p 0 divided by sigma p bar is a standard normal random variable. And the particular value that we have for this test statistic based on the random sample is equal to 0. 45 minus 0. 5 divided by 0. 022. That comes out to be minus 2. 21. Now, what is the p value in a lower tail test?

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The p value is equal to the probability that a standard normal random variable will take a value less than minus 2. 21 in this case. From the standard normal template, we find that probability to be 0. 013. Yes, and so the p value is 0. 013, which is less than our significance level of 3 percent. So we reject the null hypothesis.

Yes, that is true. The sample does provide enough evidence to claim that the proportion of people that find the arrangements inadequate is less than 50 percent. The proportion was 45 percent in the sample. And that was enough evidence to claim that the proportion is less than 50 percent in the population as well.

Let's do another example. We are going to Noida next. Dinesh Gupta works at a travel agency in Noida. As customers call in for bookings during the Diwali weekend, Dinesh claims that 90 percent of the hotels in the Noida region are fully booked for that weekend. Surprised to hear this, his manager Sangeetha calls 70 hotels and finds that 12 of them have vacancies.

She wants to determine if Dinesh's claim of 90 percent is not true. She would like us to conduct a hypothesis test using a significance level of 5%. How do we set this up and what can we conclude based on the sample data? Here Sangeetha wants to verify Dinesh's claim that 90 percent of the hotels are booked.

So what is the claim we put in as the alternate hypothesis? The alternate is going to be that the population proportion P is not equal to P0, where P0 is 0. 9. Okay. So, the null hypothesis will be the opposite statement, which is that the population proportion p is equal to p 0, where p 0 is 0. 9, and that would make this a two tailed test.

So, Sangeetha wants to use a significance level of 5 percent for the test. What is the evidence from the sample? We have 12 out of 70 hotels with vacancies. So 58 of them are booked. Therefore, n equals to 70 and the sample proportion p bar is 58 divided by 70, which is 0. 83. Yes. So next we assume that the null hypothesis is true at equality.

That is, we are indeed sampling from a population where p is equal to 0. 9. Compute the test statistic. And for that, we need the standard error sigma p bar. Since N p and N q are both more than 5, we can assume that p bar comes from a normal distribution with mean equal to the claim

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in the null hypothesis, which is 0. 9 and the standard error equal to the square root of p 0 times 1 minus p 0 divided by N. Can you calculate that? It is square root of 0. 9 multiplied by 0. 1 divided by 70, which is 0. 036. Yes, so that means our test statistic p bar minus p 0 divided by sigma p bar is a standard normal random variable and the particular value that we have for this test statistic based on the random sample is equal to 0. 83 minus 0. 9 divided by 0. 036, which comes out to be minus 1. 99.

Now, what is the p value in a two tailed test? In this case, the p value is equal to twice the probability that a standard normal random variable will take values less than minus 1. 99. From the standard normal template, we find that p value is 0. 046. Yes, and so that is less than our significance level of 5%. So we reject the null hypothesis. That is true. And the sample does provide enough evidence to refute Dinesh's claim that 90 percent of the hotels are booked. There is enough evidence to claim that the percentage is not 90%, but less than that in the population.

Now hope that gives you some comfort on hypothesis test for proportions. Let us summarize what we did for proportions. To summarize, the hypothesis test for population proportion p can be of three types, a lower tail test, an upper tail test, and a two tail test. In each case, we write the null and the alternate hypothesis.

Once you have written the statements of the null and the alternate hypothesis, then you compute the test statistic using the sample data, assuming that the null hypothesis is true at equality. The test statistic is a standard normal random variable as long as n times p zero and n times one minus p zero are both greater than or equal to five.

Then we compute the specific value of this test statistic using the sample data. Then depending on the nature of the test, lower tail, upper tail or two tail, you compute the p value. If the p value is less than or equal to alpha, then you reject the null hypothesis, otherwise you do not reject the null hypothesis. Okay.

Let us now look at hypothesis test for variance.

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Hypothesis Testing for Population Variance

As we discussed in the case of confidence intervals for population variance and standard deviation, in order to do a hypothesis test on population variance, we will use the sample variance to make inferences. In this case, the sampling distribution of the quantity n minus 1 times s square divided sigma square will be very useful.

And as we saw earlier, whenever a simple random sample of size n is selected from a normal population, the sampling distribution of n minus 1 times s square divided by sigma square is a chi square distribution with n minus 1 degrees of freedom. We will use this information to conduct hypothesis test like we did for population mean and population proportion. Let sigma 0 square be the hypothesized value of the population variance and sigma square denote the true value of the population variance. Then there are three forms of a hypothesis test for population variance. These three forms are similar to the three forms that we used to conduct the one tail and the two tail test for population mean and population proportion.

The structure is the same? Yes, like we had there. In the lower tail test, the alternate hypothesis would be of the form sigma square is less than sigma 0 square, which gives the null hypothesis as sigma square greater or equal to the hypothesized value of sigma 0 square. In the upper tail test, the alternate hypothesis would be of the form sigma square greater than sigma 0 square, which gives the null hypothesis as sigma square less than or equal to the hypothesized value of sigma 0 square.

In the two tail test, the alternate hypothesis would be of the form sigma square is not equal to sigma 0 square, which gives the null hypothesis as sigma square equal to the hypothesized value of sigma 0 square. The procedure for conducting this hypothesis test for population variance uses this hypothesized value of population variance sigma 0 square and the sample variance s square to compute the value of the test statistic.

If the population has a normal distribution, then this test statistic defined as n minus one times s square divided by sigma zero square. And that will be a chi square distribution with n minus one degrees of freedom. Yes, exactly. And after computing the value of the test statistic, the chi square test statistic, the p value approach can be used just like before.

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And if the p value is less than the significance level alpha. We reject the null hypothesis and if it is greater than the significance level, then we do not reject the null hypothesis. Let us see a couple of examples. Rakesh Adhikari oversees a bus service in Meerut. He would like to ensure that the arrival times at the bus stops have low variability.

He wants to guarantee that the arrival times have a variance of four minutes or less from the schedule times. To verify whether his claims are being met. He conducts a random sample of 24 bus arrivals at the Abu Lane market area in the city. He finds the sample variance S square to be 4. 9 minutes.

Assuming that the population distribution of arrival times is approximately normal, he wants to know whether the sample provides enough evidence to support or reject his claim using a significance level of 5%. Ashwini, want to set up the null and alternate hypothesis? Who is trying to make the claim? And what are they trying to claim from the sample data?

Rakesh is wanting claim that the variance is 4 minutes or less. Yes, and so the alternate hypothesis is going to be that the population variance sigma square is less than sigma 0 square which is 4 minutes. And now the null hypothesis will be the opposite statement which is that the population variance sigma square is greater than or equal to sigma zero square, which is four minutes.

And that would make this an upper tail test? Yes, and so Rakesh has asked us to use a significance level of five percent for this test. What is the evidence from our sample? We have s square of 4. 9 minutes from a sample of n equals to 24. Yes, so now we assume that the null hypothesis is true at equality.

That is, we are indeed sampling from a normal population with sigma square equal to 4 minutes or less. And we compute the test statistic as n minus 1 times s square divided by sigma 0 square, which comes out to be 23 times 4. 9 divided by 4, which comes out to be 28. 18. Now, what is the p value in an upper tail test?

The p value is equal to the probability that a chi square with 23 degrees of freedom will take values more than 28. 18. From the chi square template, we find that probability to be 0. 209.

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Yes, so the p value is 0. 209, which is more than our significance level of 5%. So we cannot reject the null hypothesis.

That is true. The sample does not provide us enough evidence to claim that the variance is more than 4 minutes. Like we did with the population means and proportions, we can write other forms of the hypothesis test for population variance as well. Let us look at a two tailed test related to population variance.

Ram Singh works at the Motor Vehicles Department in Gorakhpur. Historically, the variance in test scores of the individuals applying for driver's license at the Motor Vehicles Department in Gorakhpur has been 100. However, Ram Singh has just introduced a new exam with a new set of test questions. So, he and his colleagues at the motor vehicles department would like the variance of the new exam to remain at the historical level.

To verify whether the variance has changed or not, Ram Singh collects a sample of 30 exam scores and he finds that the sample variance S square is 162. Using a significance level of 5 percent, he would like to know whether this evidence supports a claim that the new exam has a variance in the test score that is different from the historical value.

So Tejas, help us get going. What are the null and the alternate hypothesis? Who's trying to make the claim? And what are they trying to claim using the sample data? Ram Singh is wanting claim that the variance is not equal to 100. Yes. So the alternate is going to be The population variance sigma square is not equal to sigma 0 square, which is 100.

And the null hypothesis will therefore be the opposite statement, which is that the population variance sigma square is equal to sigma 0 square, which is equal to 100. Now Rakesh has asked us to use a significance level of 5 percent for this test. What is the evidence from the sample? We have s square of 162 from a sample of n equals to 30.

Yes, so now we assume that the null hypothesis is true at equality. That is, we are indeed sampling from a normal population with sigma square equal to 100. Now we compute the test statistic as n minus 1 times s square divided by sigma 0 square, which comes out to be 29 times 162 divided by 100, which comes out to be equal to 46. 98. Now, what is the p value in a two

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tailed test? The p value is equal to two times the upper tail probability corresponding to 46. 98 in a chi square with 29 degrees of freedom.

From the chi square template, we find that probability to be 0. 037. Yes, so the p value is 0. 037, which is less than our significance level of 5%. So we reject the null hypothesis. Yes, that is true. The sample provides enough evidence to say that the new exam has a different variance in the test scores.

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Summary of Hypothesis Testing for Population Variance

Let us summarize the steps related to the hypothesis test for variance. To summarize, the hypothesis test for population variance sigma square can be of three types, a lower tail test, an upper tail test, and a two tail test, with sigma 0 square as the hypothesized value of the population variance at each test.

In each case, we write the null and the alternate hypothesis. Once you have written the statements of the null and the alternate hypothesis, then you compute the test statistic using the sample data, assuming that the null hypothesis is true at equality. The test statistic in this case is n minus 1 times s square divided by sigma 0 square, which is a chi square random variable with n minus 1 degrees of freedom as long as we are sampling from a normal population. We compute the specific value of the test statistic using the sample data. Then, depending on the nature of the test, lower tail, upper tail, or two tail, you compute the p value. If the p value is less than or equal to alpha, then you reject the null hypothesis.

Otherwise, you do not reject the null hypothesis. Now, that we have done several examples of hypothesis test for population mean mu, population proportion p and the population variance sigma square. Let us summarize before we close this module. In this module, we built on the foundations that we had established for sampling distributions of mean, proportion and variance to make a leap into statistical inference, where we are using sample data to make inferences about the population. We did that through two important techniques. The first was confidence intervals and the second was hypothesis tests.

With respect to confidence intervals, we were able to use the point estimates obtained from the sample to provide interval ranges within which the population mean mu or the population proportion P or the population variance sigma square are likely to lie. For the case of the population mean and proportion, these intervals were written as the sample point estimate plus or minus a margin of error. In the case of the population variance, we were able to directly write the population variance to lie between an lower limit and an upper limit with a certain level of confidence.

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We also saw how to set sample sizes that would allow us to obtain a desired margin of error in the case of population mean and population proportion. Next we looked at hypothesis testing, which allows us to make claims about a population parameter. Claims such as whether a population parameter is equal to a certain value, greater than a certain value or less than a certain value.

We did this for population mean mu, population proportion p, and population variance sigma squared. In doing the hypothesis test, we need to be careful about how to structure the test, particularly while writing down the null and the alternate hypothesis. It always helps to see who is trying to make the claim about the population parameter, and what is the specific claim that they are trying to make.

Once the hypothesis test is set up and the significance value is determined, then we assume that the null hypothesis is true at equality to compute the test statistic. We note the random variable that the test statistic corresponds to. Sometimes it has a t distribution, sometimes the standard normal distribution, and sometimes it has the chi square distribution.

And then, we compute the p value, the probability of making a type 1 error. And if the probability of making a type 1 error is small enough, then we are able to reject the null hypothesis. Otherwise, we do not reject the null hypothesis. Both confidence intervals and hypothesis tests have a lot of applications in making statistical inferences from a single random sample. That is obtained from the population.

In the next module, we will continue our journey into statistical inference by looking at another important technique known as regression. So, thank you Ashwini and Tejas for all your help and support and thank all of you for your attention. I will see you next time.

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